

## EEE 263 – ELECTRONIC CIRCUITS

# Bipolar Junction Transistor

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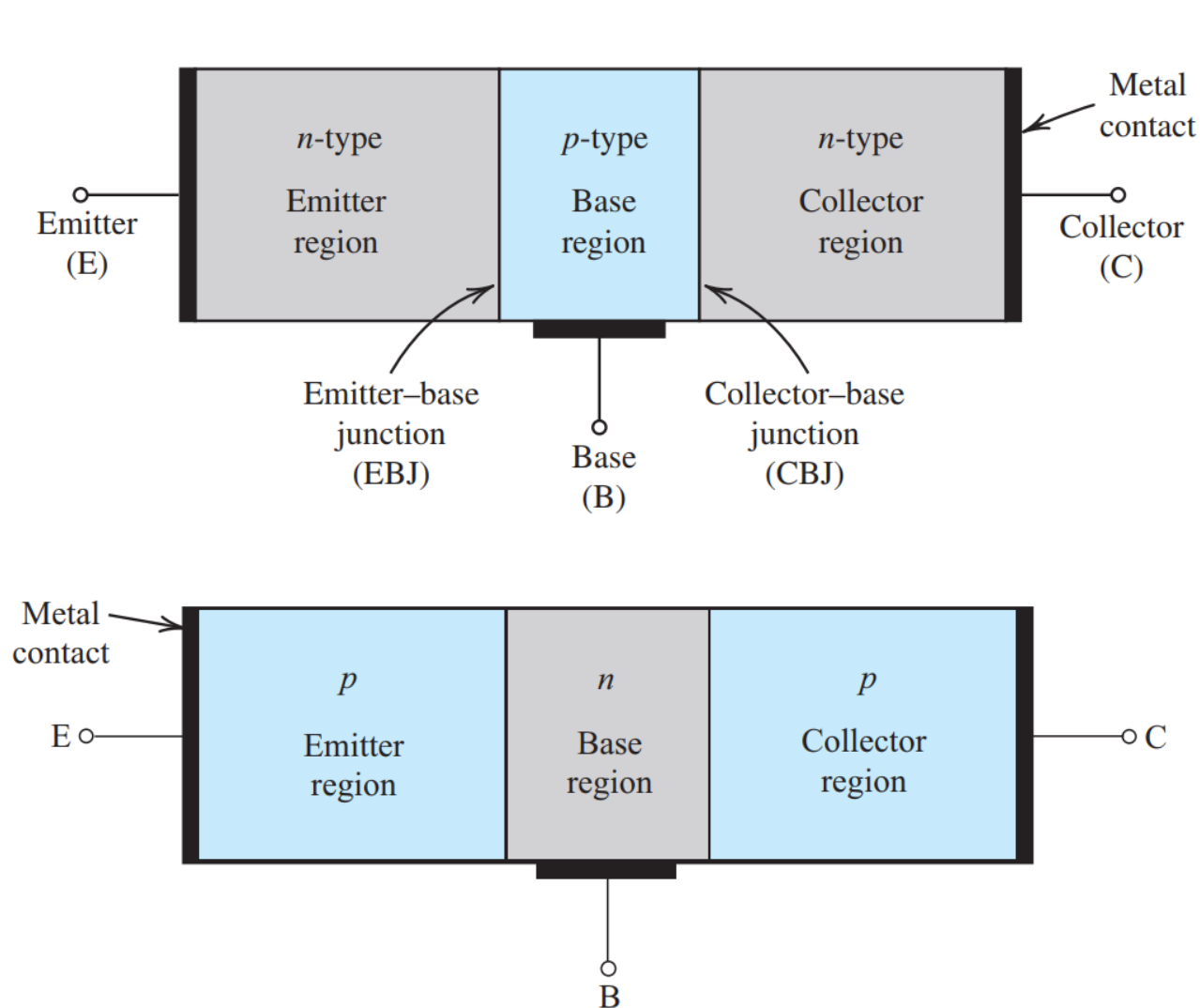
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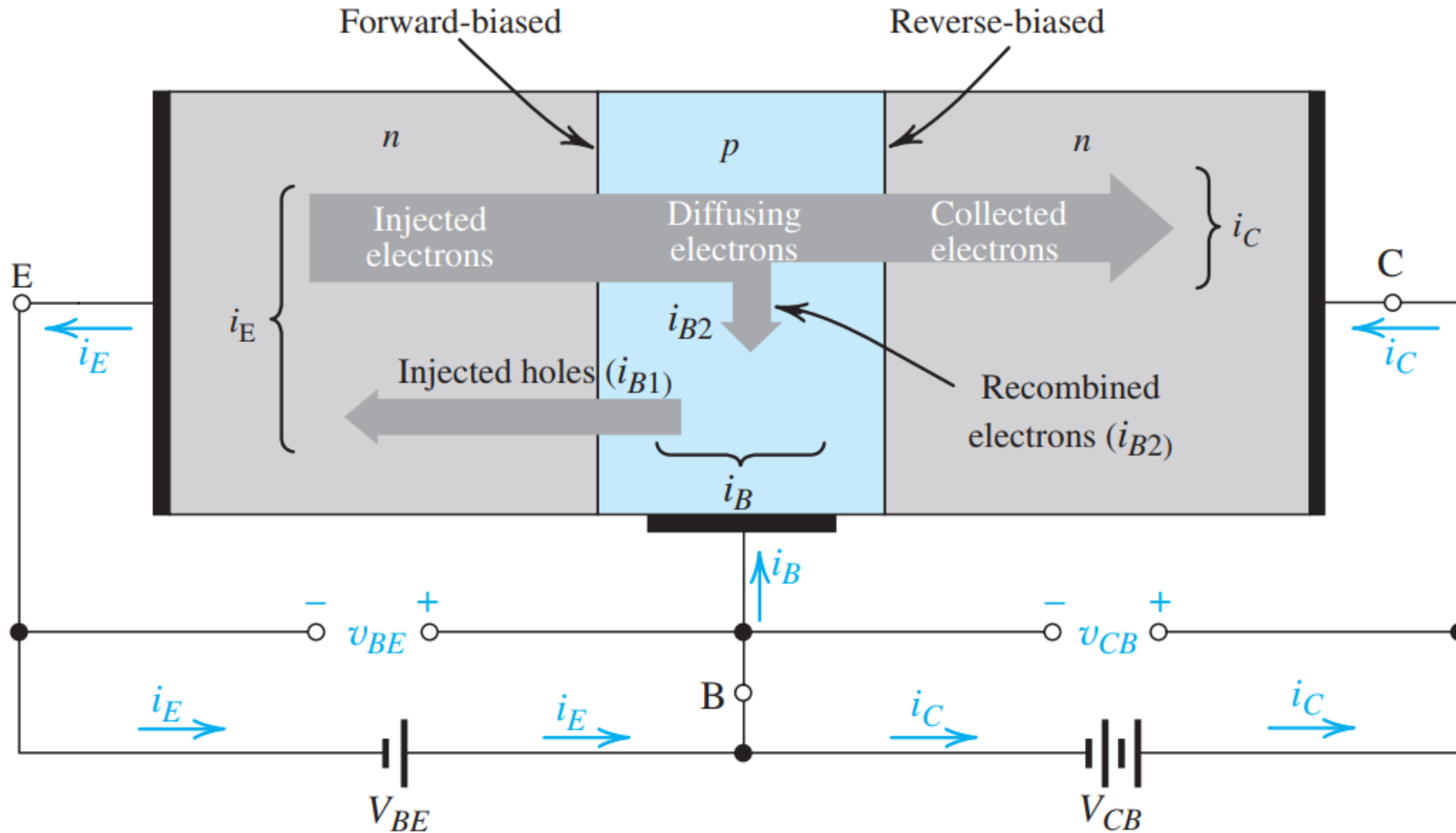
[khairul.buet.ac.bd](http://khairul.buet.ac.bd)

# Bipolar Junction Transistor



| Table 6.1 BJT Modes of Operation |         |         |
|----------------------------------|---------|---------|
| Mode                             | EBJ     | CBJ     |
| Cutoff                           | Reverse | Reverse |
| Active                           | Forward | Reverse |
| Saturation                       | Forward | Forward |

# Operation of the npn Transistor in the Active Mode



$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta}$$

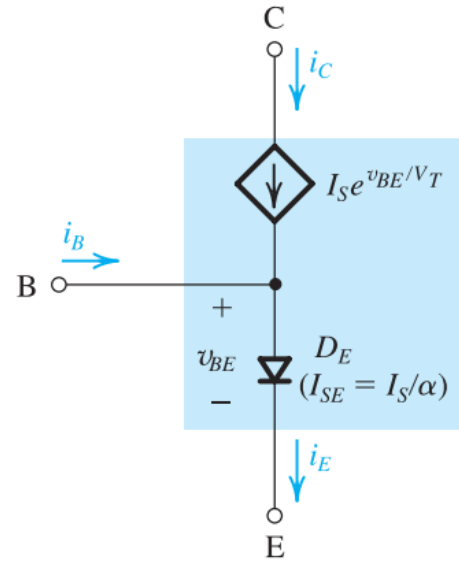
$$i_B = \left( \frac{I_S}{\beta} \right) e^{v_{BE}/V_T}$$

$$i_E = i_C + i_B$$

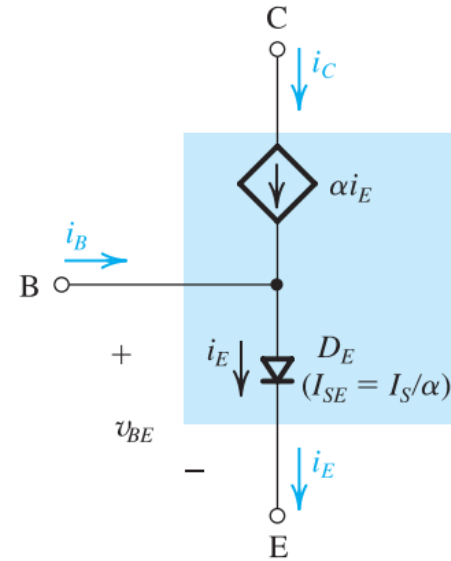
$$i_C = \alpha i_E$$

$$\alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{1 - \alpha}$$

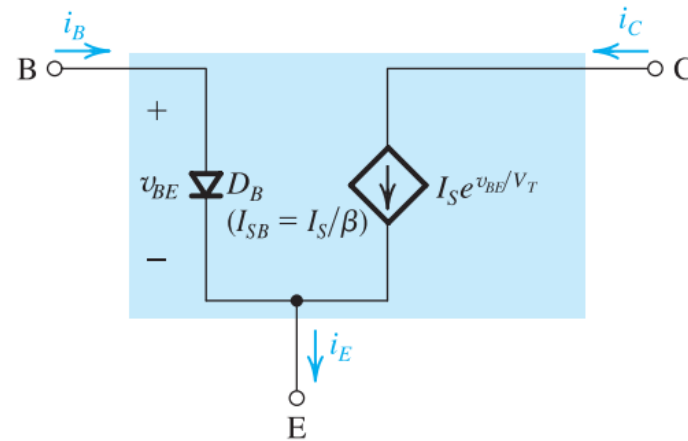
# Equivalent Models in Active Mode



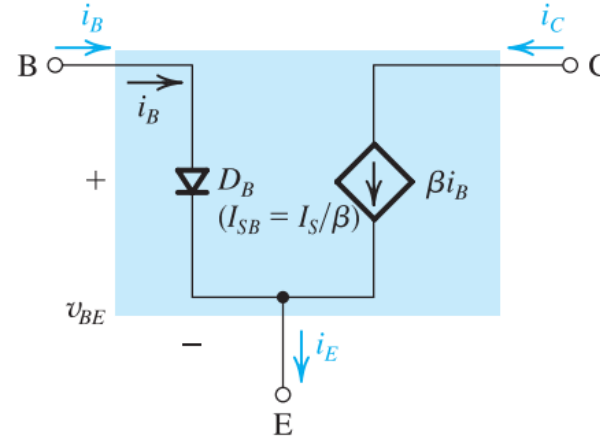
(a)



(b)

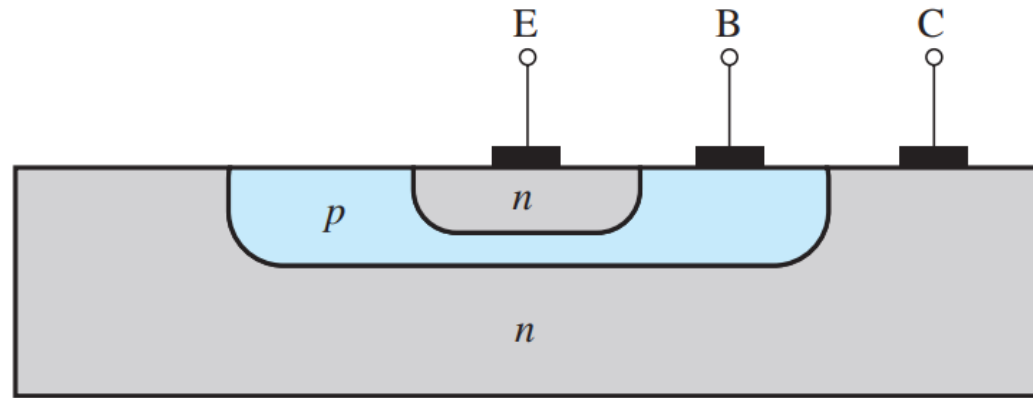


(c)

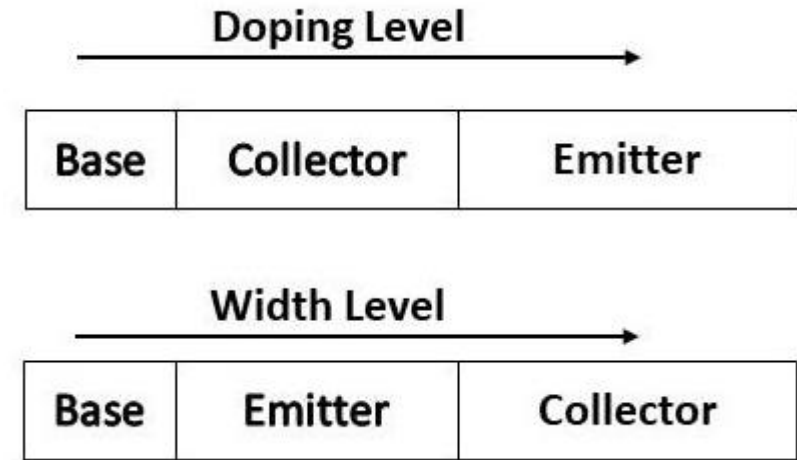


(d)

# Structure of Actual Transistors

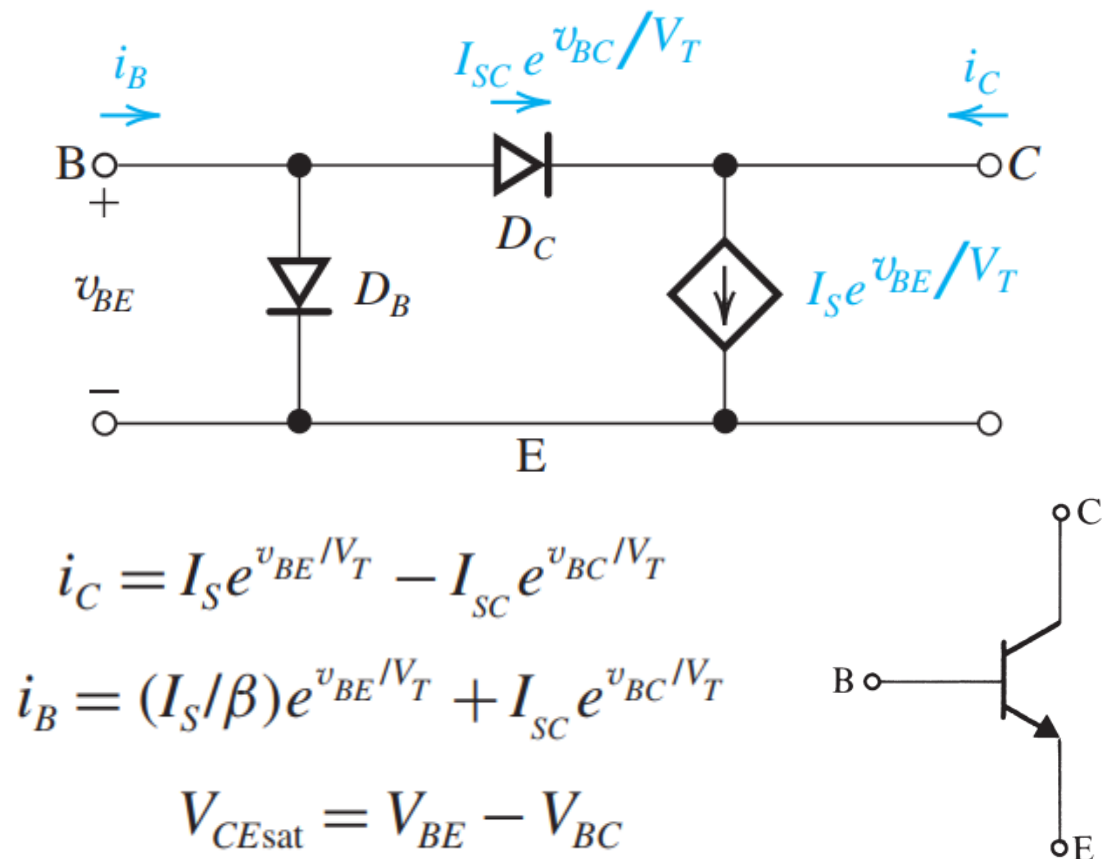


Structure of Actual Transistors



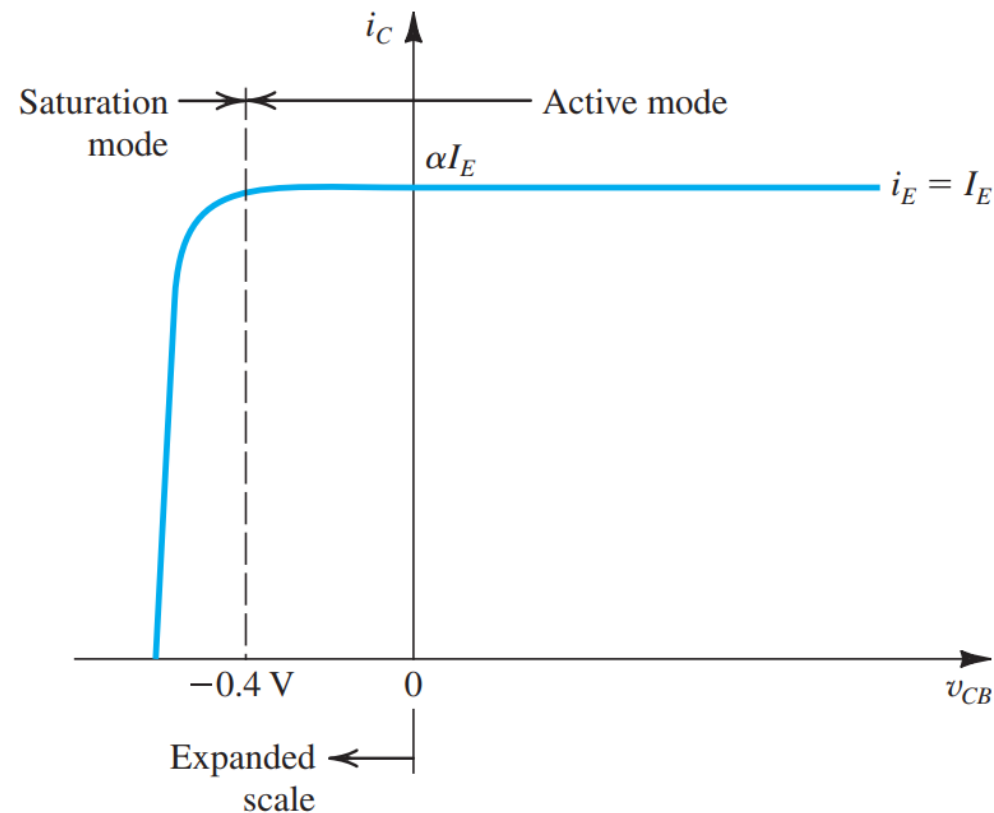
- Observe that the device is **not symmetrical**, and thus the **emitter and collector cannot be interchanged**. The MOSFET, on the other hand, being a perfectly symmetrical device, can operate equally well with its drain and source terminals interchanged.
- If the emitter and collector are reversed—that is, the CBJ is forward biased and the EBJ is reverse biased—the device operates in a mode called the “**reverse-active mode**.” The resulting values of  $\alpha$  and  $\beta$ , denoted  $\alpha_R$  and  $\beta_R$  (with R denoting reverse), are much lower than the values of  $\alpha$  and  $\beta$ , respectively, obtained in the “forward”-active mode discussed above. Hence, **the reverse-active mode has no practical application**.

# Operation in the Saturation Mode



CBJ has a much larger area than the EBJ,  $V_{BC}$  will be smaller than  $V_{BE}$  by 0.1 to 0.3V. Thus,

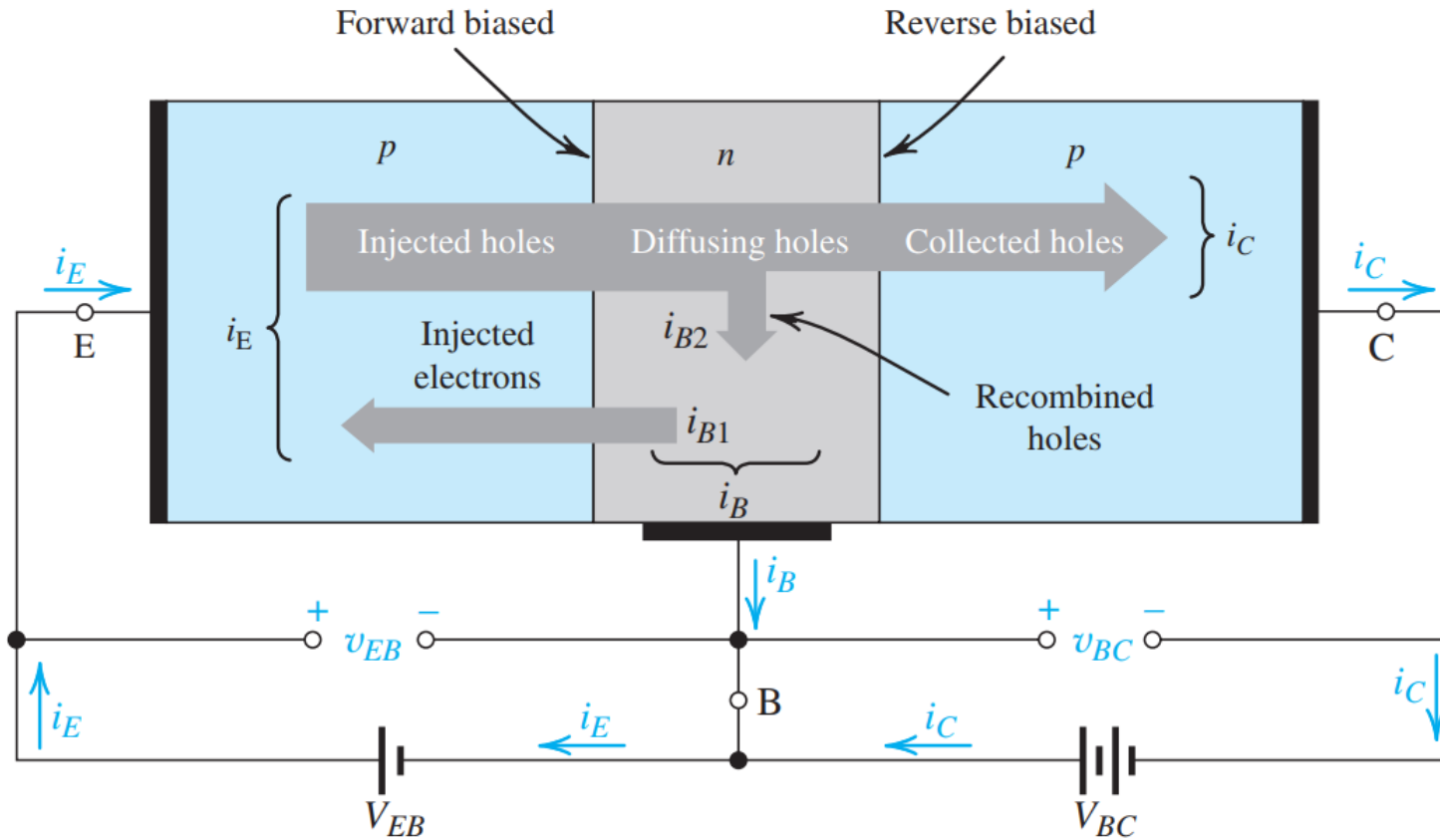
$$V_{CEsat} \simeq 0.1 \text{ to } 0.3 \text{ V}$$



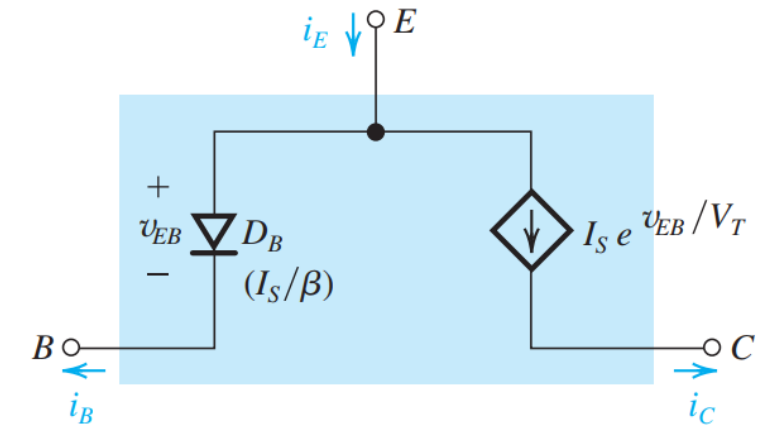
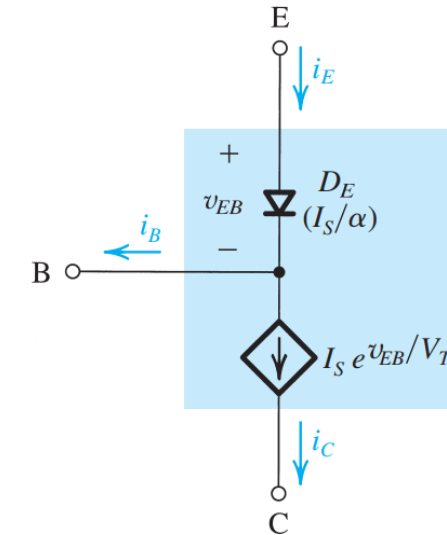
**Typically,**

- **At the edge of saturation:  $V_{CEsat} = 0.3 \text{ V}$**
- **In deep saturation:  $V_{CEsat} = 0.2 \text{ V}$**

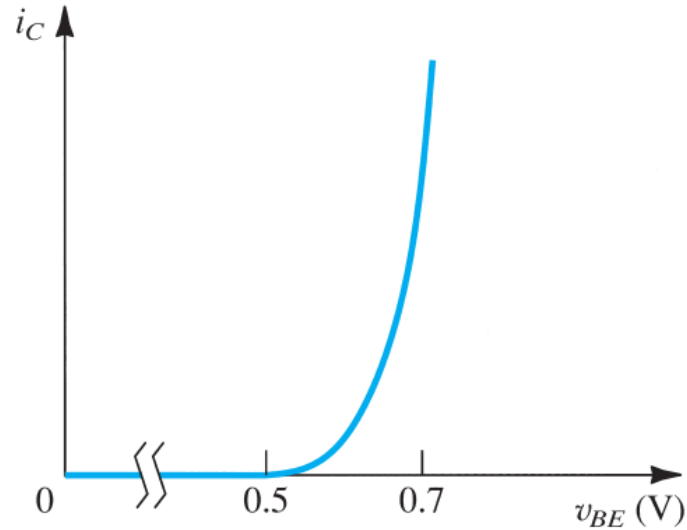
# The pnp Transistor



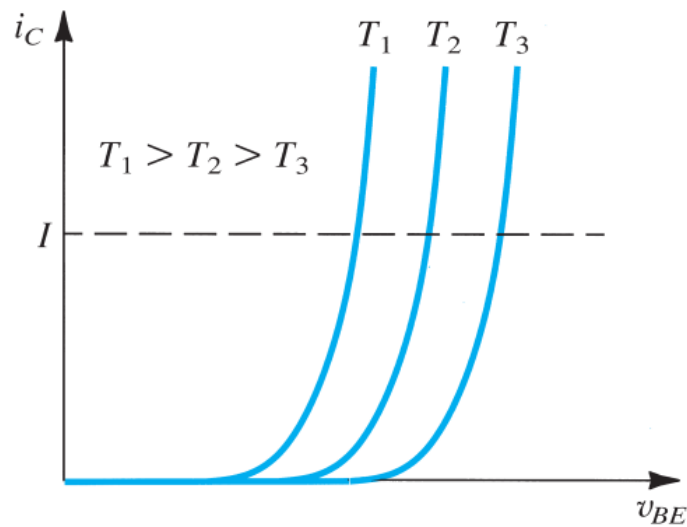
Current flow in a *pnp* transistor biased to operate in the active mode.



# Current-Voltage Characteristics



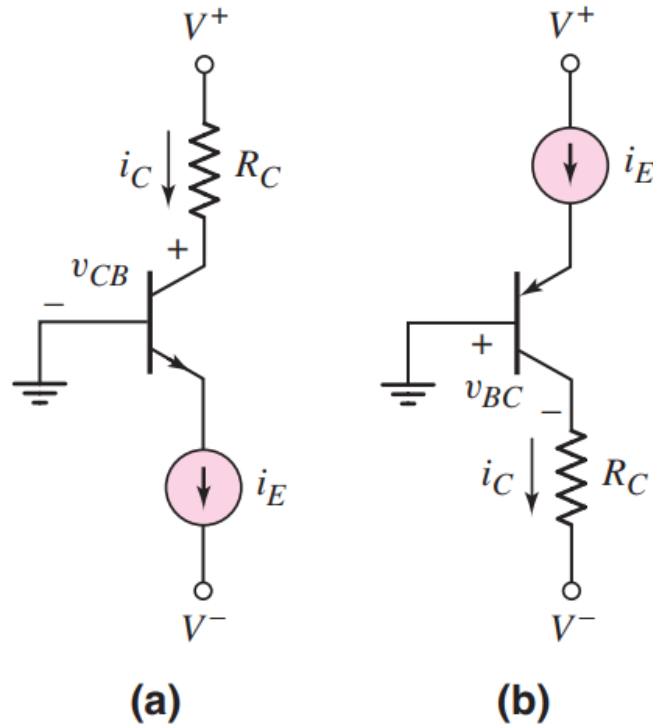
**Figure 6.16** The  $i_C$ - $v_{BE}$  characteristic for an *npn* transistor.



**Figure 6.17** Effect of temperature on the  $i_C$ - $v_{BE}$  characteristic. At a constant emitter current (broken line),  $v_{BE}$  changes by  $-2$  mV/ $^{\circ}$ C.

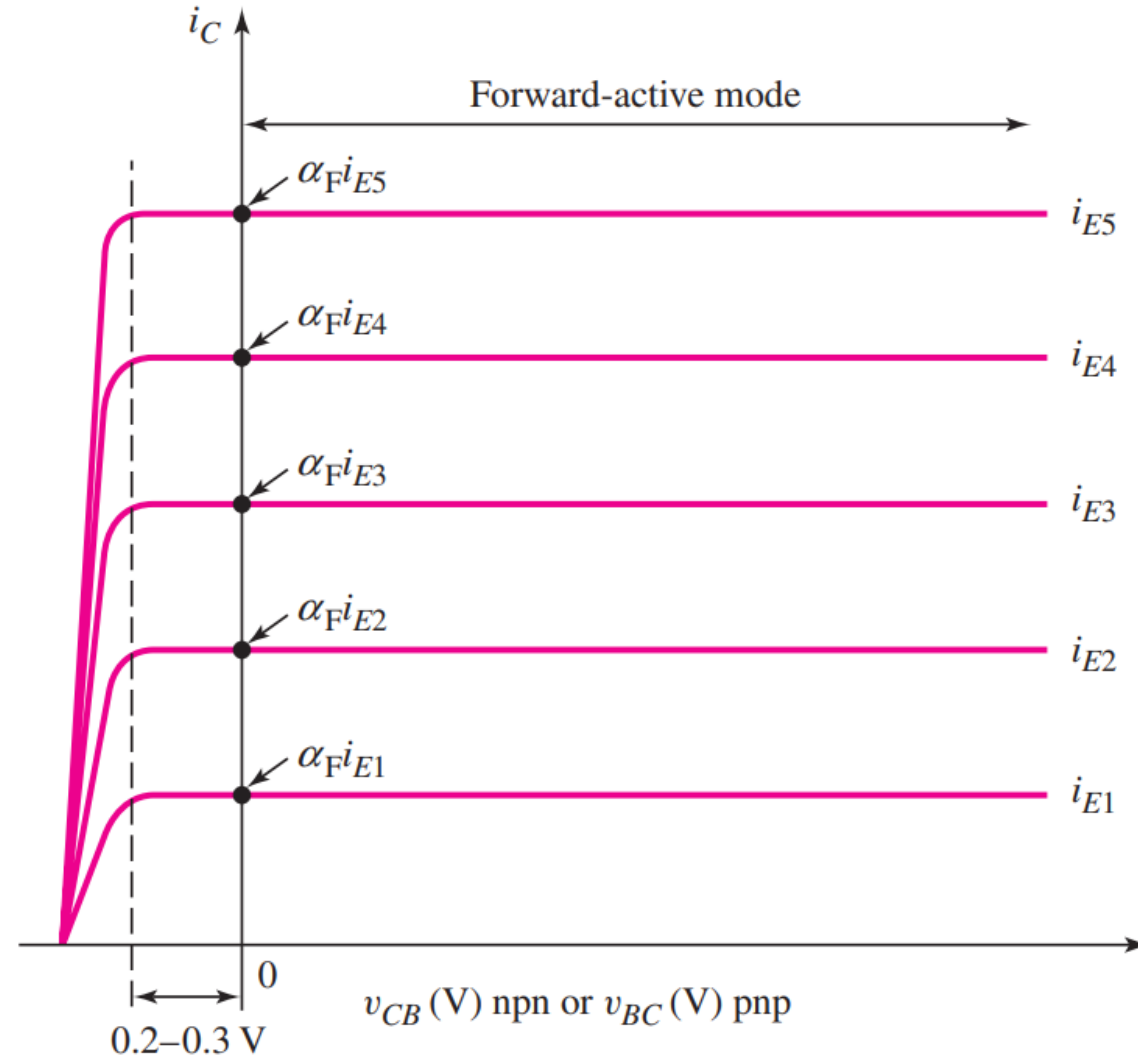


# Current-Voltage Characteristics



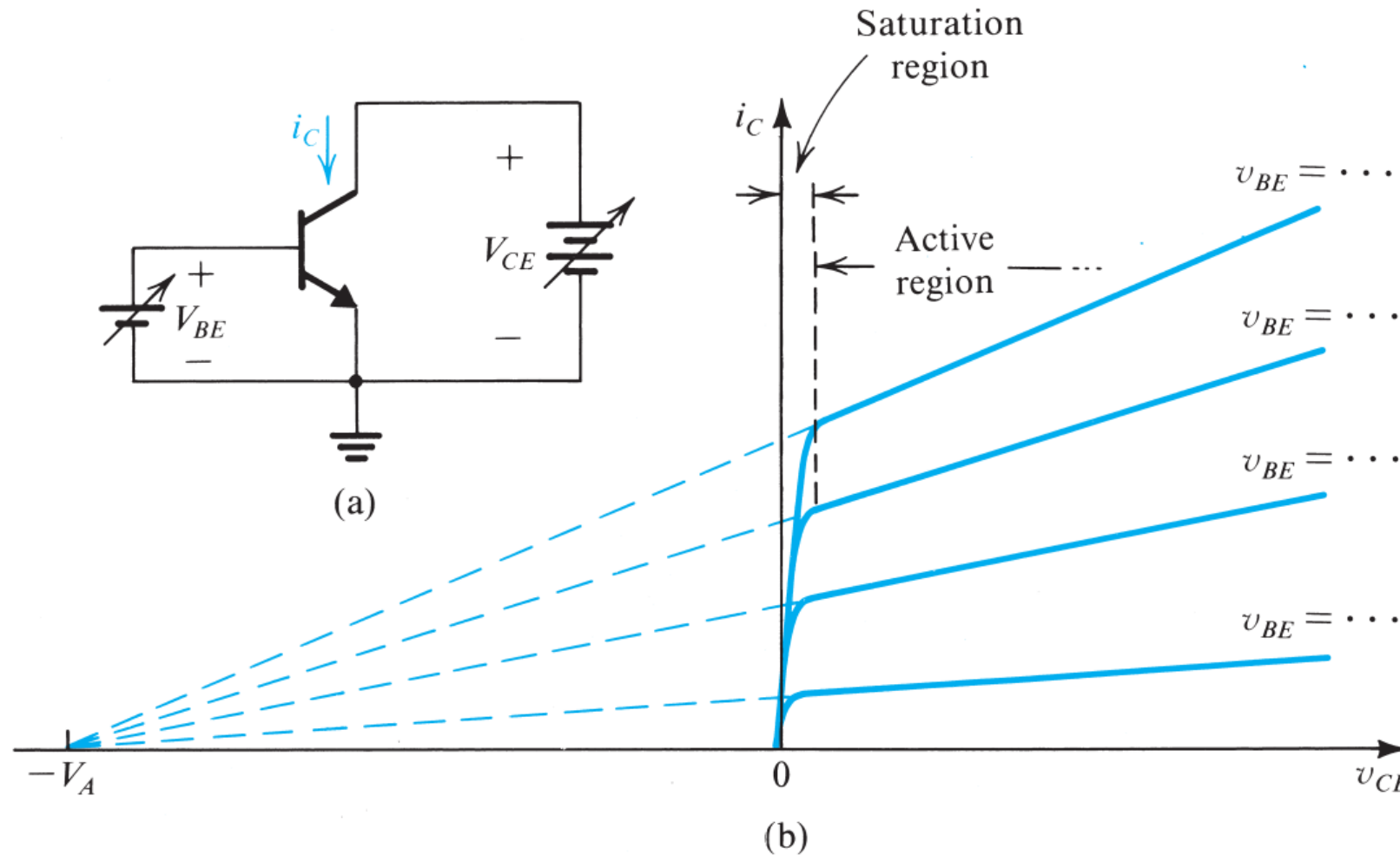
Common-base circuit configuration with constant current source biasing:

- (a) an npn transistor and
- (b) a pnp transistor



Transistor current–voltage characteristics of the common-base circuit

# Current-Voltage Characteristics

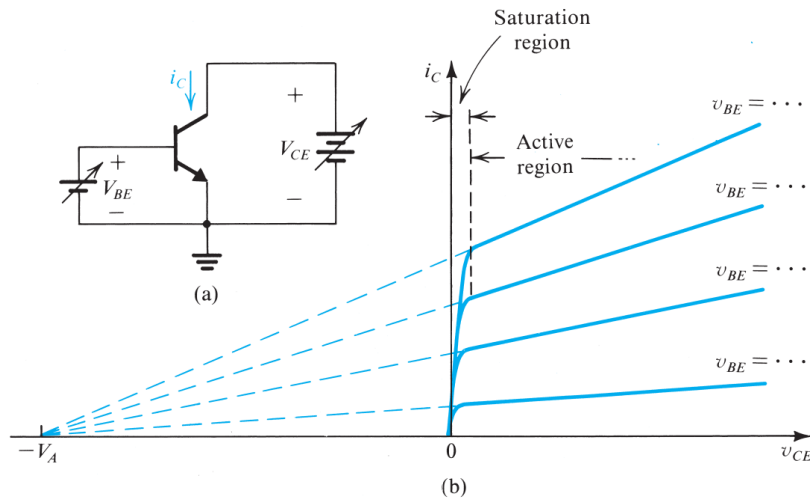


$$i_C = I_S e^{v_{BE}/V_T}$$

$$I_S = \frac{A_E q D_n n_i^2}{N_A W}$$

**Figure 6.18** (a) Conceptual circuit for measuring the  $i_C$ - $v_{CE}$  characteristics of the BJT. (b) The  $i_C$ - $v_{CE}$  characteristics of a practical BJT.

# Current-Voltage Characteristics



**Figure 6.18** (a) Conceptual circuit for measuring the  $i_C$ - $v_{CE}$  characteristics of the BJT. (b) The  $i_C$ - $v_{CE}$  characteristics of a practical BJT.

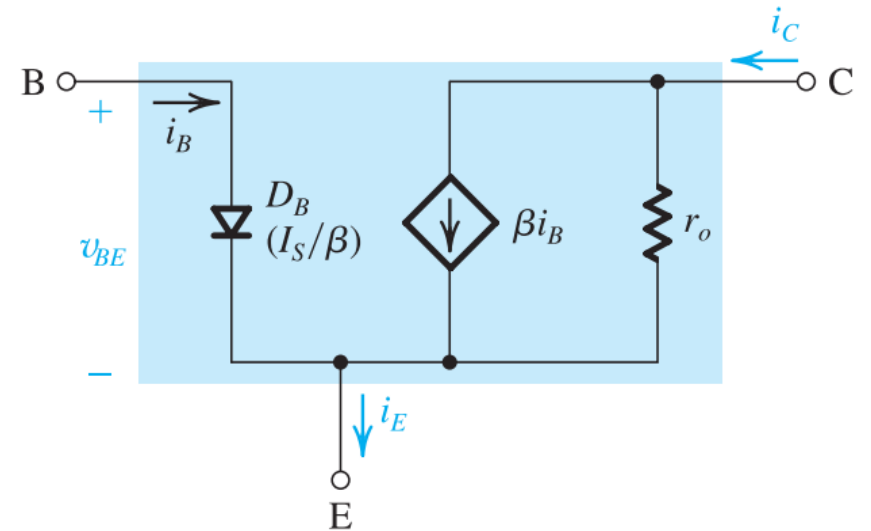
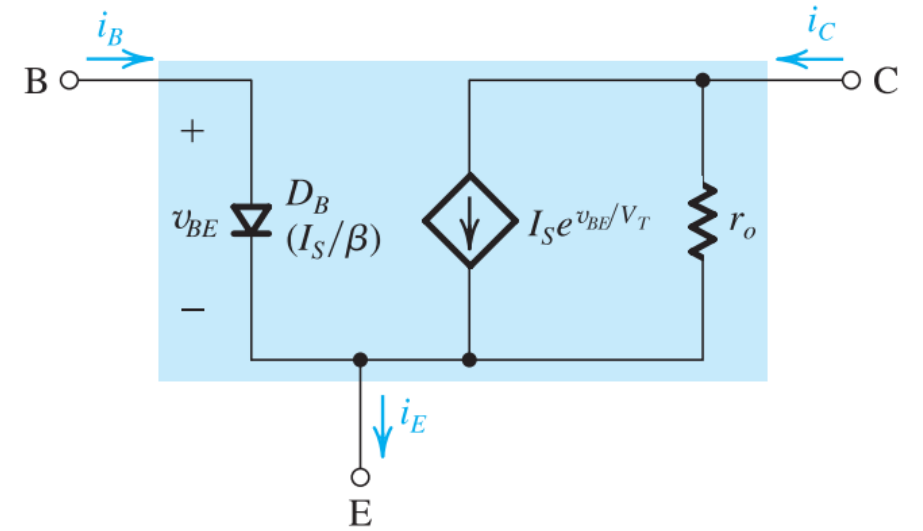
$$i_C = I_S e^{v_{BE}/V_T} \left( 1 + \frac{v_{CE}}{V_A} \right)$$

$$r_o \equiv \left[ \frac{\partial i_C}{\partial v_{CE}} \bigg|_{v_{BE} = \text{constant}} \right]^{-1}$$

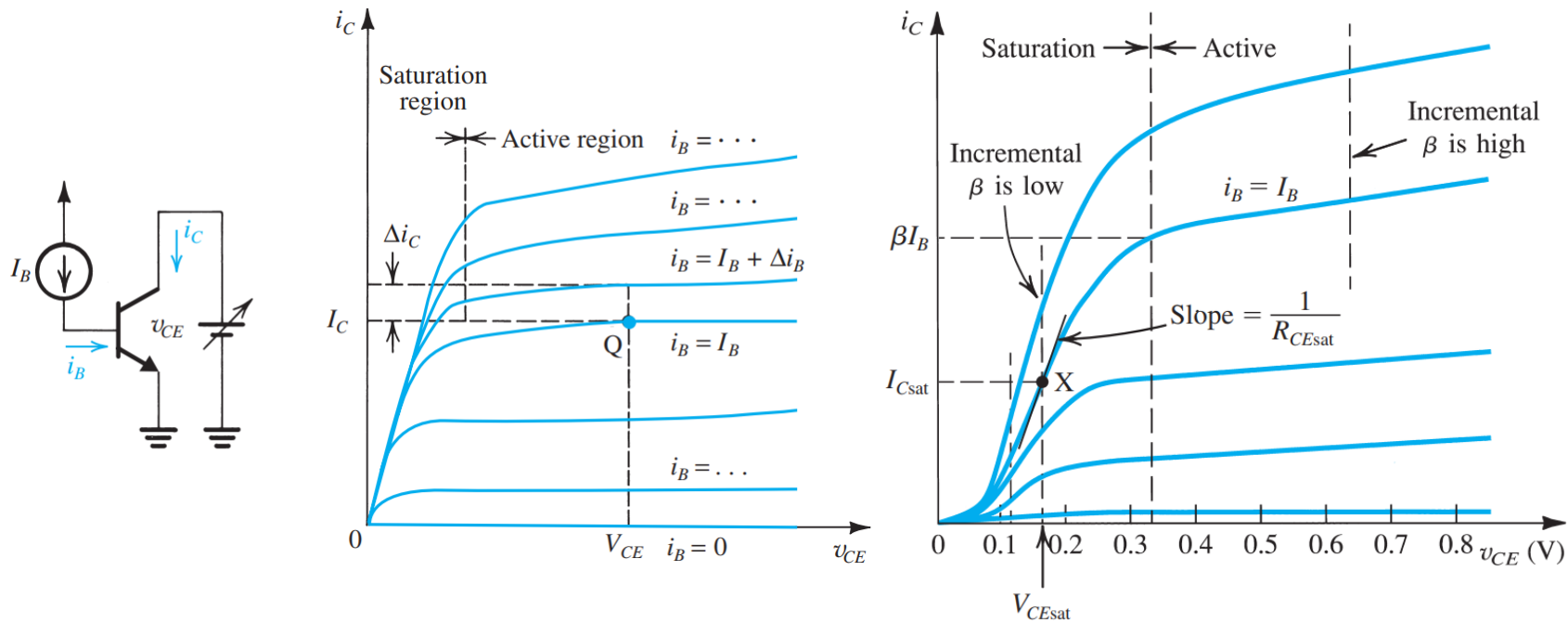
$$r_o = \frac{V_A + V_{CE}}{I_C}$$

$$I'_C = I_S e^{V_{BE}/V_T}$$

$$r_o = \frac{V_A}{I'_C}$$

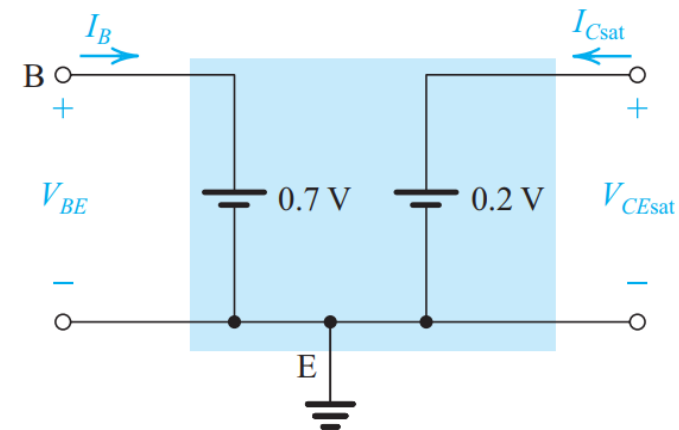


# Current-Voltage Characteristics



$$I_{Csat} = \beta_{\text{forced}} I_B, \text{ where } \beta_{\text{forced}} < \beta$$

$$R_{CEsat} \equiv \left. \frac{\partial v_{CE}}{\partial i_C} \right|_{\substack{i_B = I_B \\ i_C = I_{Csat}}}$$



# Current-Voltage Characteristics

## Summary of the BJT Current-Voltage Relationships in the Active Mode

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left( \frac{I_S}{\beta} \right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left( \frac{I_S}{\alpha} \right) e^{v_{BE}/V_T}$$

Note: For the *pnp* transistor, replace  $v_{BE}$  with  $v_{EB}$ .

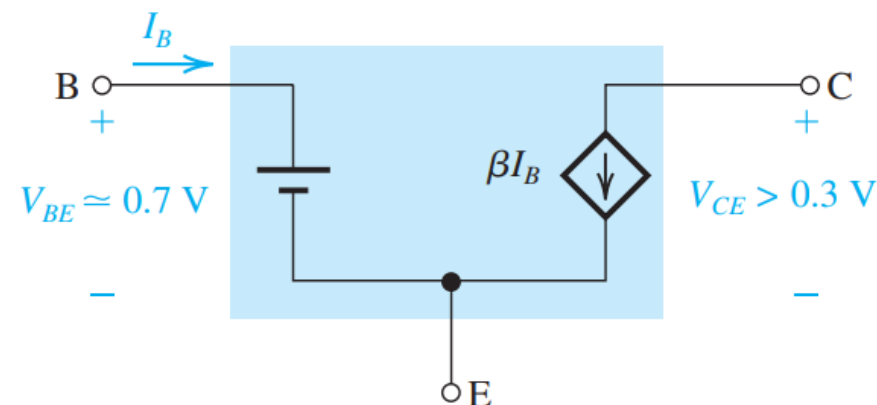
$$i_C = \alpha i_E \qquad i_B = (1 - \alpha) i_E = \frac{i_E}{\beta + 1}$$

$$i_C = \beta i_B \qquad i_E = (\beta + 1) i_B$$

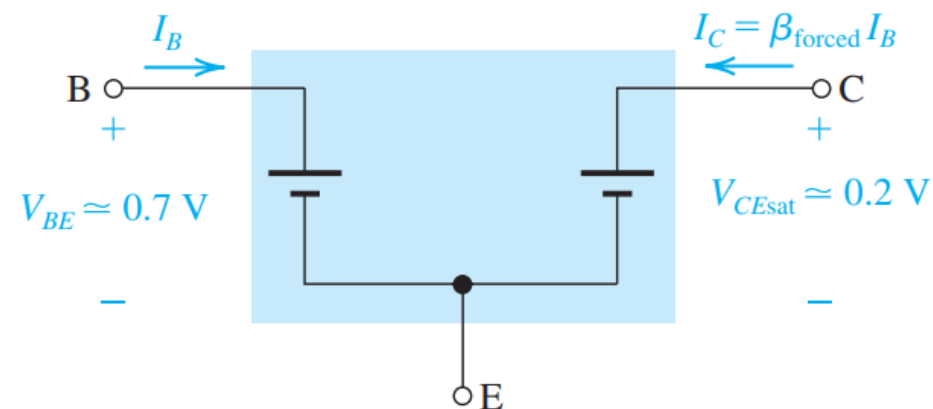
$$\beta = \frac{\alpha}{1 - \alpha} \qquad \alpha = \frac{\beta}{\beta + 1}$$

$$V_T = \text{thermal voltage} = \frac{kT}{q} \simeq 25 \text{ mV at room temperature}$$

**Active**  
EBJ:  
Forward  
Biased  
  
CBJ:  
Reverse  
Biased



**Saturation**  
EBJ:  
Forward  
Biased  
  
CBJ:  
Forward  
Biased



# BJT Circuits at DC

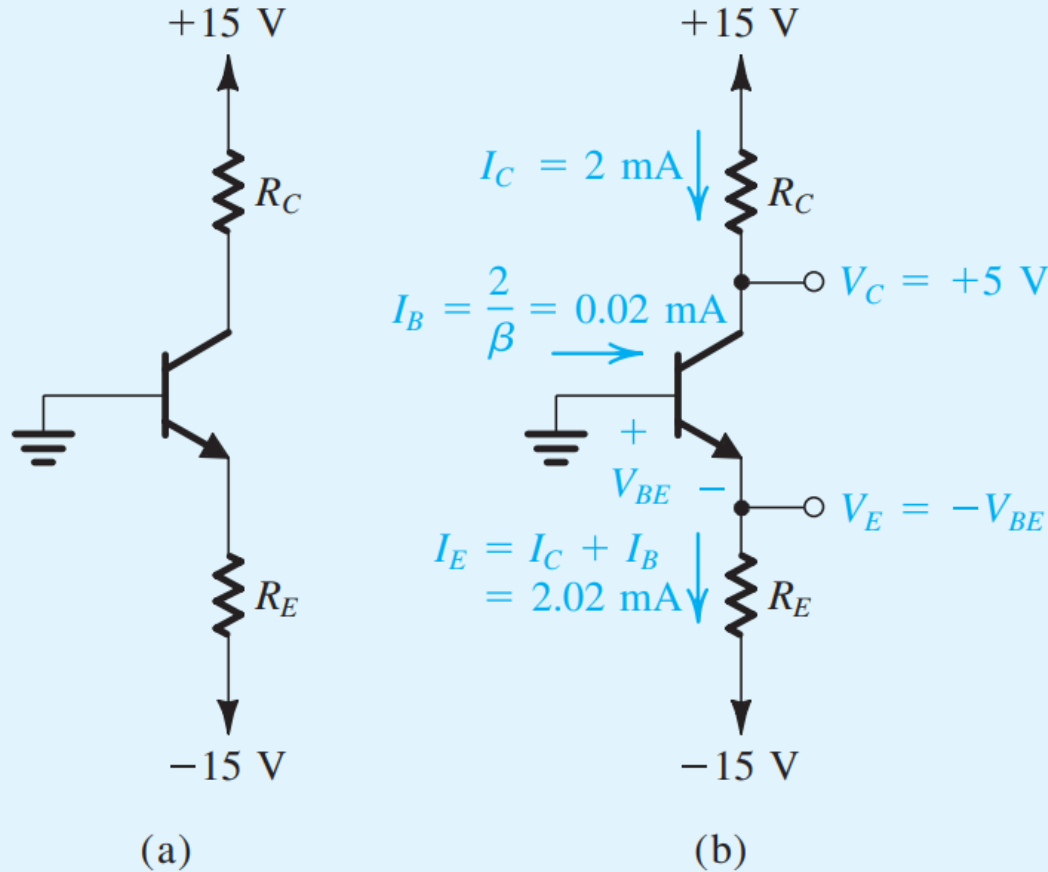
An *npn* transistor having  $I_S = 10^{-15}$  A and  $\beta = 100$  is connected as follows: The emitter is grounded, the base is fed with a constant-current source supplying a dc current of  $10 \mu\text{A}$ , and the collector is connected to a 5-V dc supply via a resistance  $R_C$  of  $3 \text{ k}\Omega$ . Assuming that the transistor is operating in the active mode, find  $V_{BE}$  and  $V_{CE}$ . Use these values to verify active-mode operation. Replace the current source with a resistance connected from the base to the 5-V dc supply. What resistance value is needed to result in the same operating conditions?

## Solution

$$\begin{aligned} V_{BE} &= V_T \ln \frac{I_B}{I_S/\beta} & V_{CE} &= V_{CC} - R_C I_C & R_B &= \frac{V_{CC} - V_{BE}}{I_B} \\ &= 25 \ln \left( \frac{10 \times 10^{-6}}{10^{-17}} \right) & I_C &= \beta I_B = 100 \times 10 \times 10^{-6} = 10^{-3} \text{ A} = 1 \text{ mA} & &= \frac{5 - 0.69}{10 \mu\text{A}} = 431 \text{ k}\Omega \\ &= 690 \text{ mV} = 0.69 \text{ V} & V_{CE} &= 5 - 3 \times 1 = +2 \text{ V} \end{aligned}$$

# BJT Circuits at DC

The transistor in the circuit of Fig. 6.15(a) has  $\beta = 100$  and exhibits a  $v_{BE}$  of 0.7 V at  $i_C = 1$  mA. Design the circuit so that a current of 2 mA flows through the collector and a voltage of +5 V appears at the collector.



$$R_C = \frac{10 \text{ V}}{2 \text{ mA}} = 5 \text{ k}\Omega$$
$$V_{BE} = 0.7 + V_T \ln\left(\frac{2}{1}\right) = 0.717 \text{ V}$$
$$V_E = -0.717 \text{ V}$$
$$I_E = \frac{I_C}{\alpha} = \frac{2}{0.99} = 2.02 \text{ mA}$$
$$R_E = \frac{V_E - (-15)}{I_E} = \frac{-0.717 + 15}{2.02} = 7.07 \text{ k}\Omega$$

Figure 6.15 Circuit for Example 6.2.

# BJT Circuits at DC

In the circuit shown in Fig. E6.13, the voltage at the emitter was measured and found to be  $-0.7$  V. If  $\beta = 50$ , find  $I_E$ ,  $I_B$ ,  $I_C$ , and  $V_C$ .

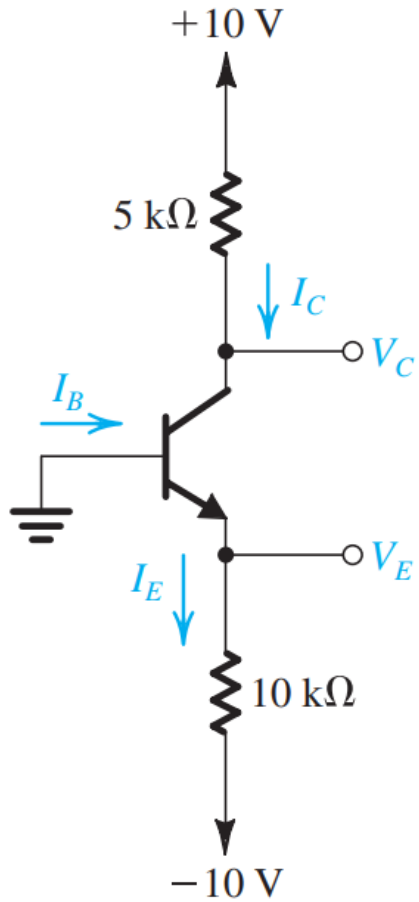


Figure E6.13

Ans.  $0.93\text{ mA}$ ;  $18.2\text{ }\mu\text{A}$ ;  $0.91\text{ mA}$ ;  $+5.45\text{ V}$



# BJT Circuits at DC

In the circuit shown in Fig. E6.14, measurement indicates  $V_B$  to be +1.0 V and  $V_E$  to be +1.7 V. What are  $\alpha$  and  $\beta$  for this transistor? What voltage  $V_C$  do you expect at the collector?

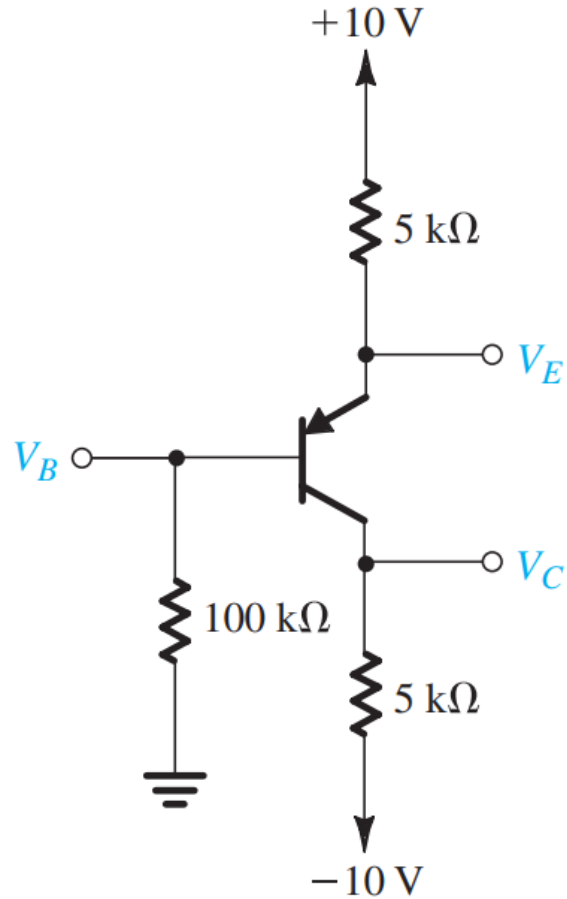


Figure E6.14

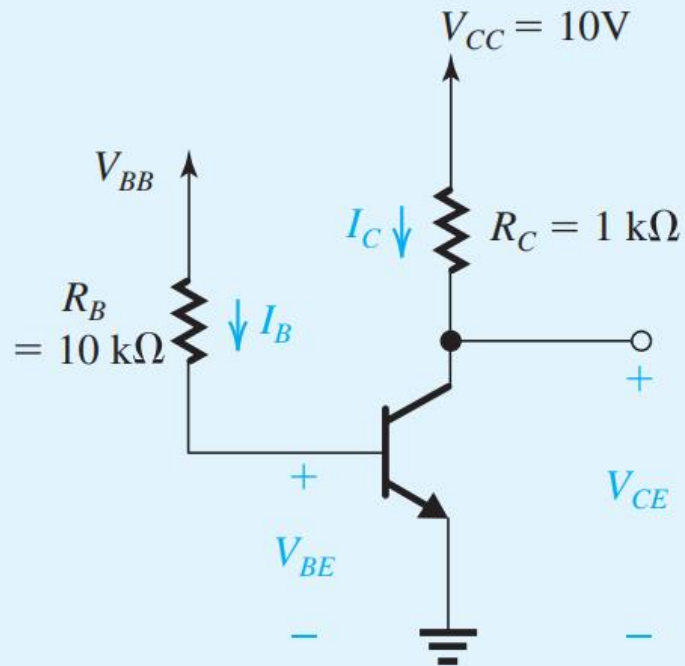
Ans. 0.994; 165; -1.75 V

# BJT Circuits at DC

For the circuit in Fig. 6.22, it is required to determine the value of the voltage  $V_{BB}$  that results in the transistor operating

- (a) in the active mode with  $V_{CE} = 5\text{ V}$
- (b) at the edge of saturation
- (c) deep in saturation with  $\beta_{\text{forced}} = 10$

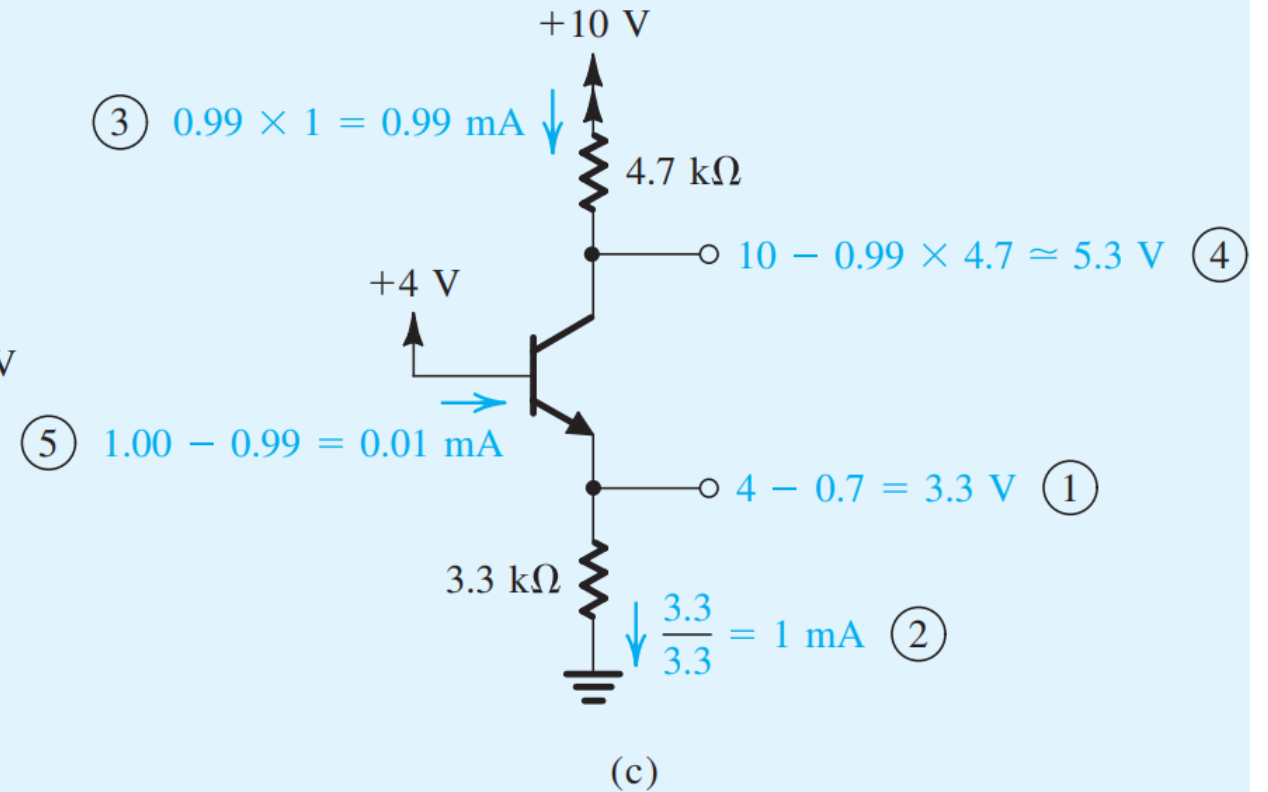
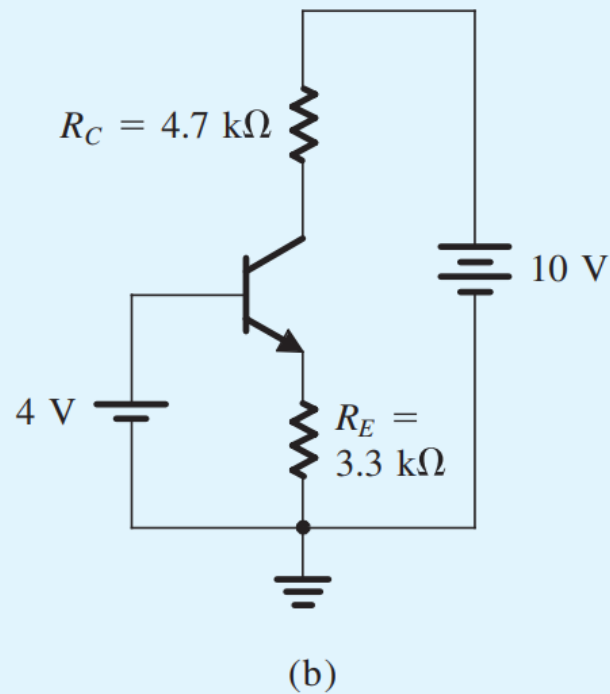
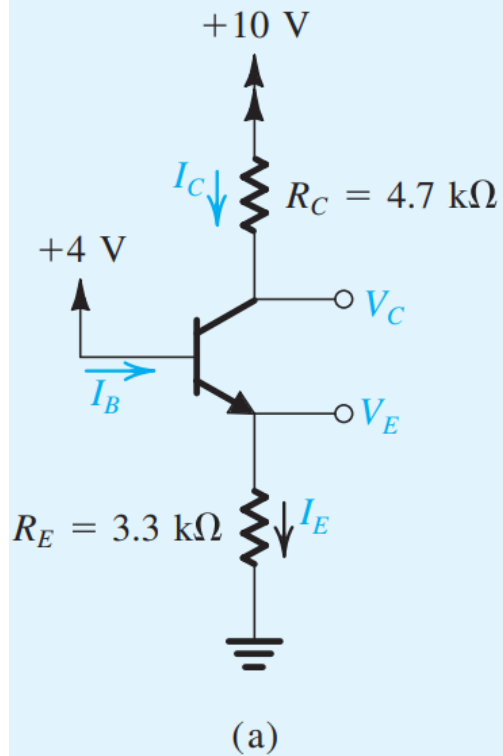
For simplicity, assume that  $V_{BE}$  remains constant at 0.7 V. The transistor  $\beta$  is specified to be 50.



**Figure 6.22** Circuit for Example 6.3.

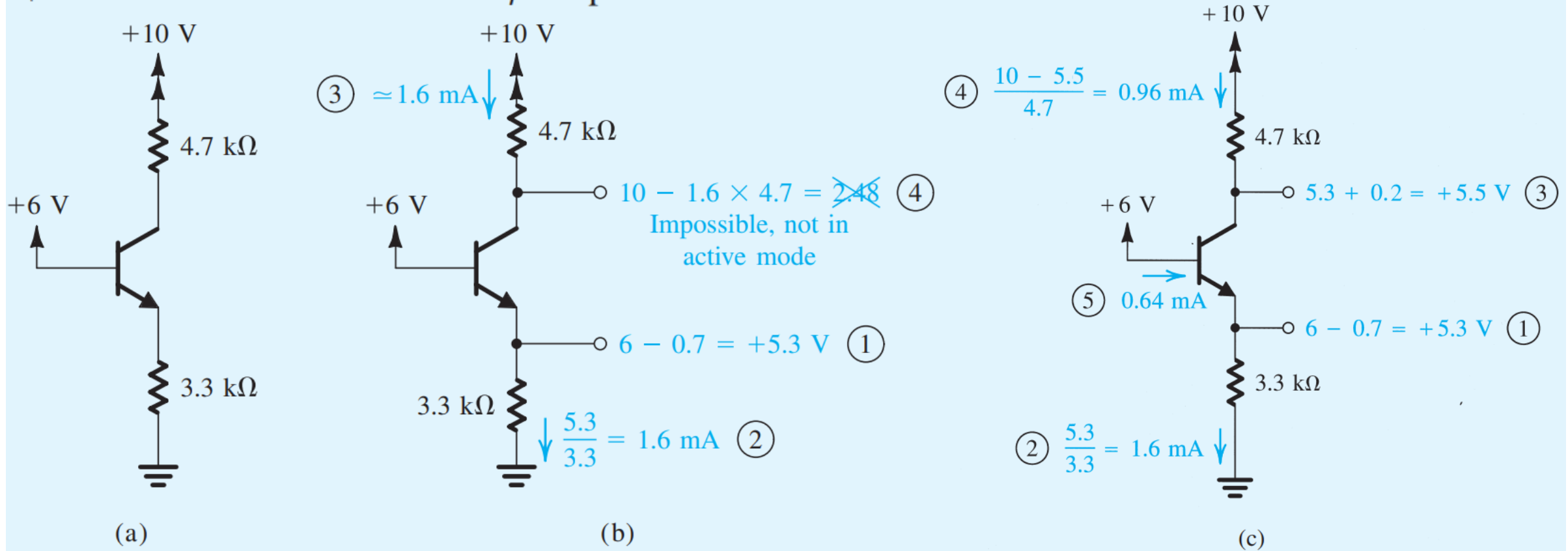
# BJT Circuits at DC

Consider the circuit shown in Fig. 6.23(a), which is redrawn in Fig. 6.23(b) to remind the reader of the convention employed throughout this book for indicating connections to dc sources. We wish to analyze this circuit to determine all node voltages and branch currents. We will assume that  $\beta$  is specified to be 100.



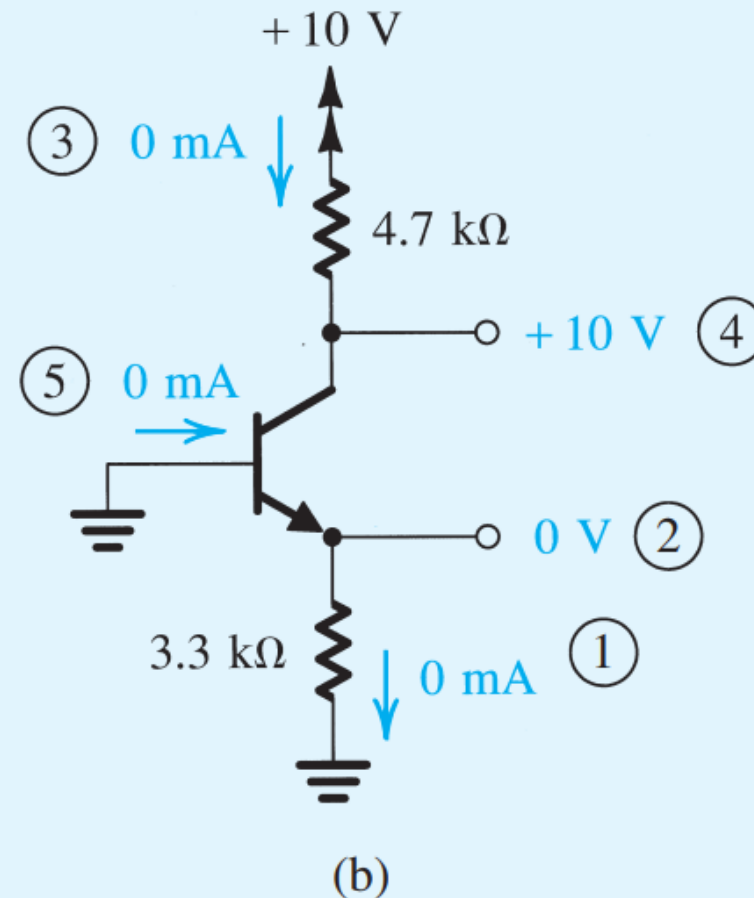
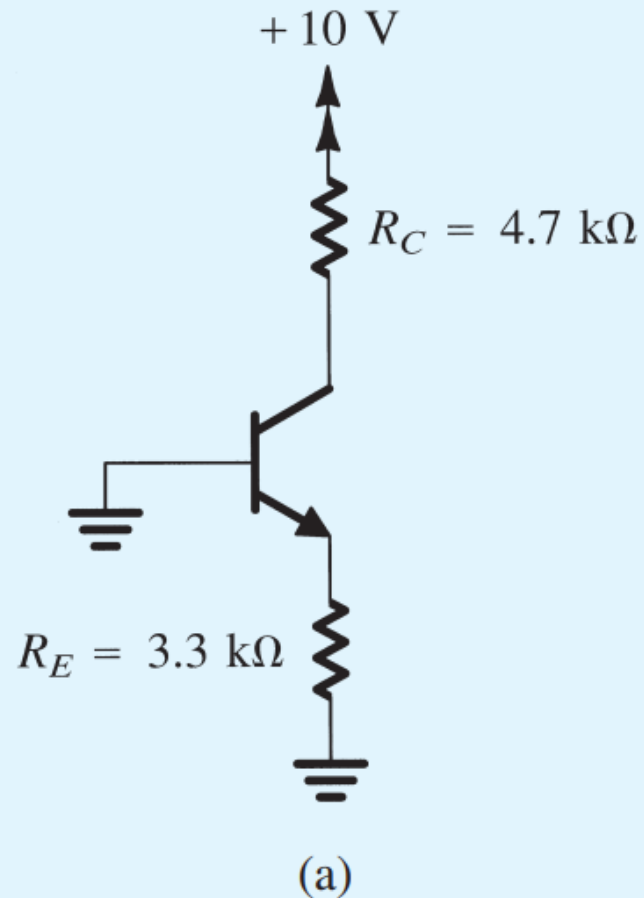
# BJT Circuits at DC

We wish to analyze the circuit of Fig. 6.24(a) to determine the voltages at all nodes and the currents through all branches. Note that this circuit is identical to that of Fig. 6.23 except that the voltage at the base is now +6 V. Assume that the transistor  $\beta$  is specified to be *at least* 50.



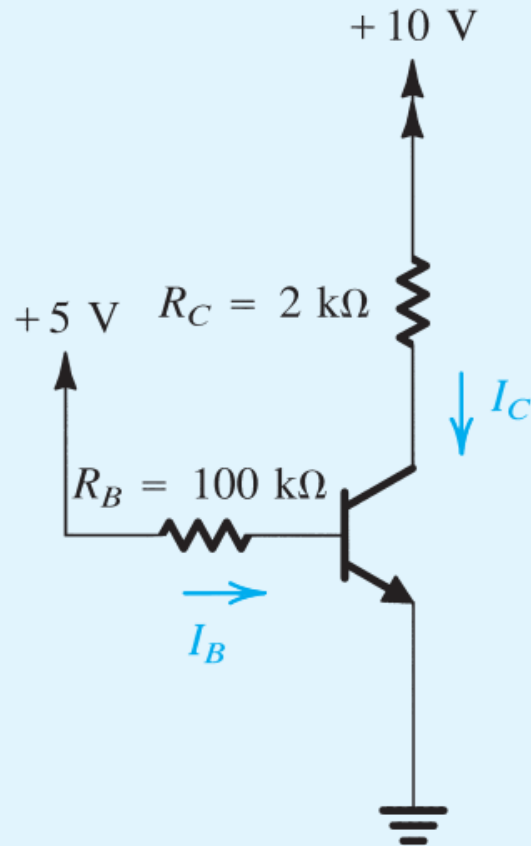
# BJT Circuits at DC

We wish to analyze the circuit in Fig. 6.25(a) to determine the voltages at all nodes and the currents through all branches. Note that this circuit is identical to that considered in Examples 6.4 and 6.5 except that now the base voltage is zero.

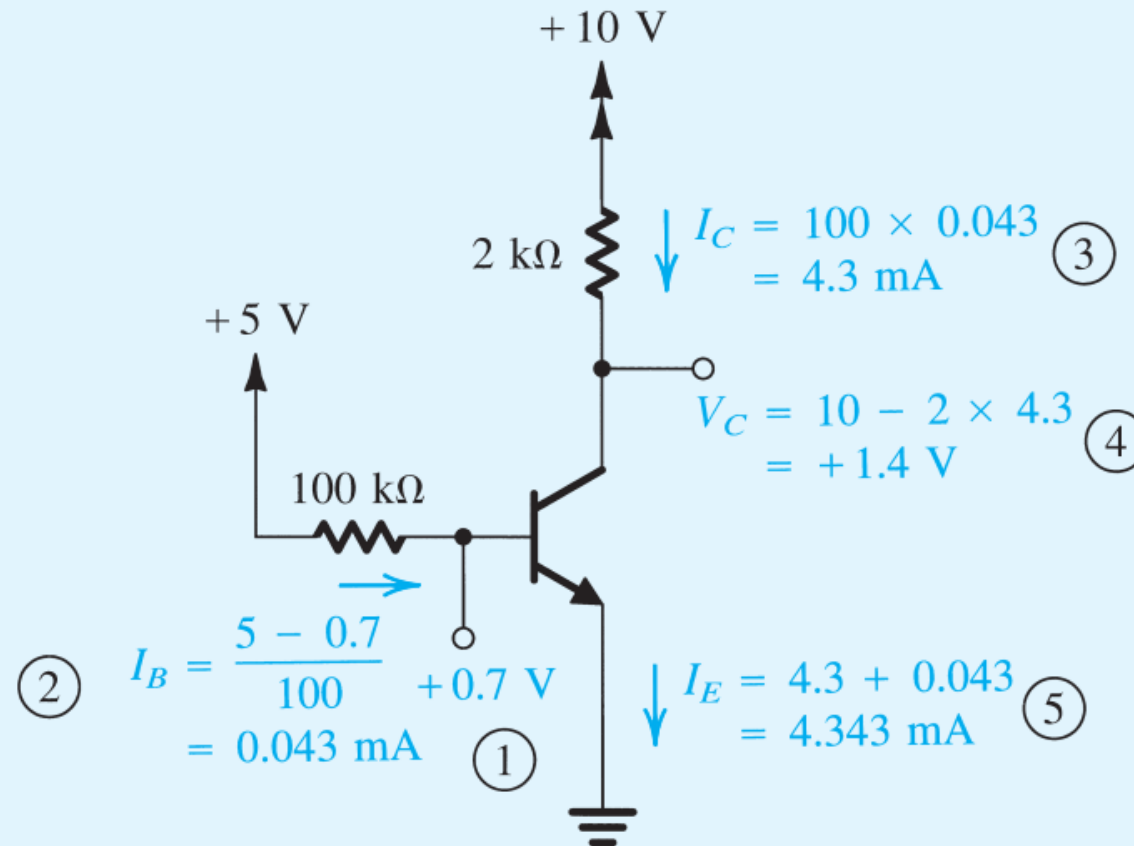


# BJT Circuits at DC

We want to analyze the circuit in Fig. 6.27(a) to determine the voltages at all nodes and the currents in all branches. Assume  $\beta = 100$ .



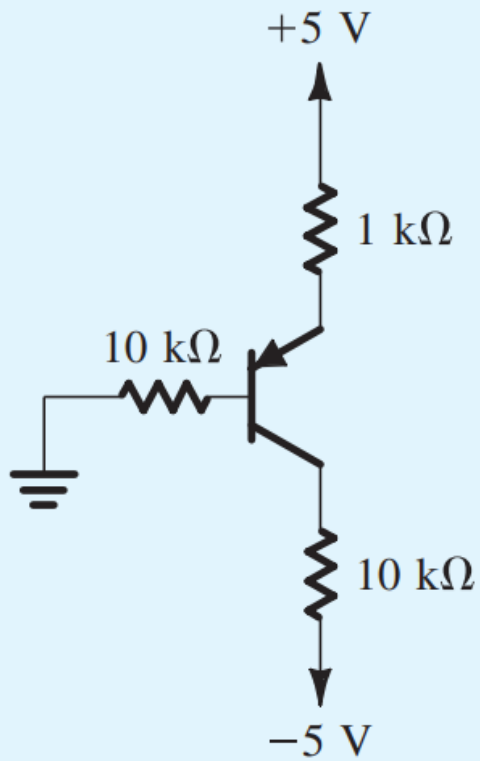
(a)



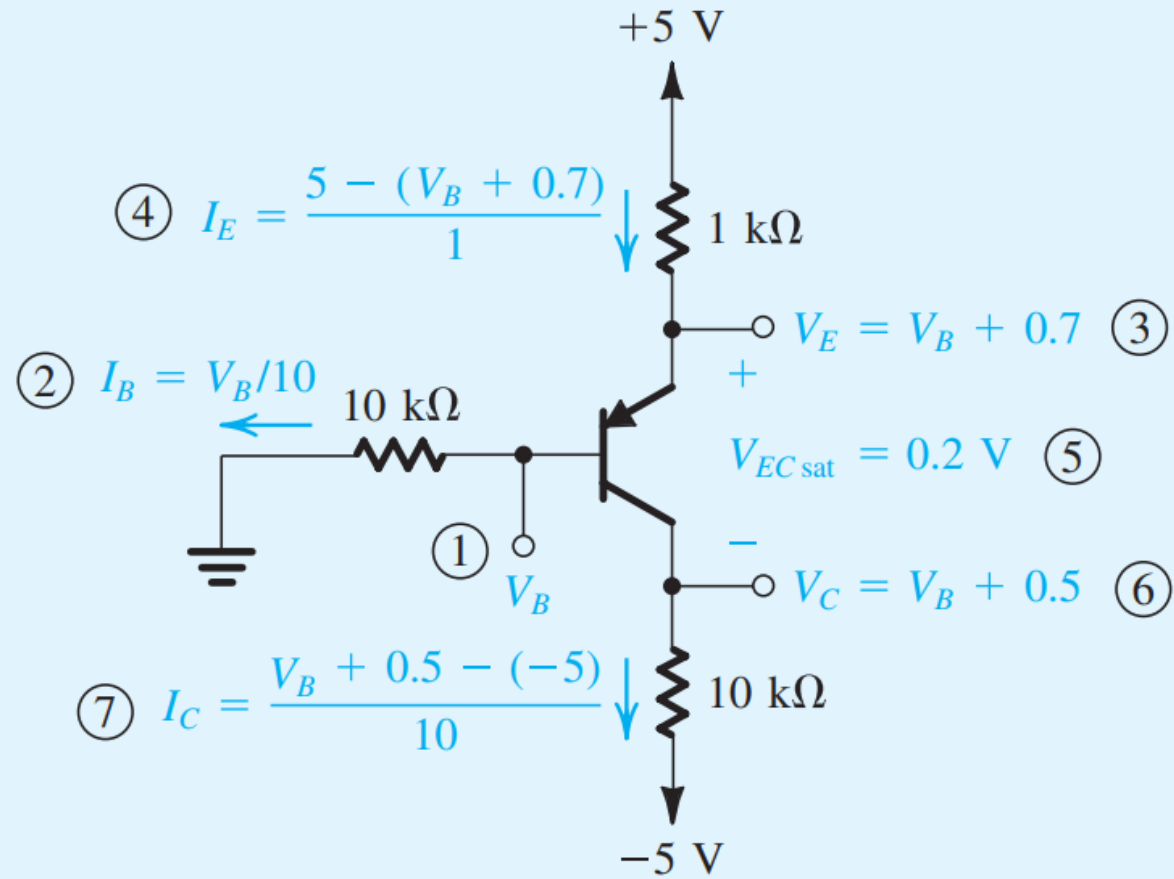
(b)

# BJT Circuits at DC

We want to analyze the circuit of Fig. 6.28(a) to determine the voltages at all nodes and the currents through all branches. The minimum value of  $\beta$  is specified to be 30.



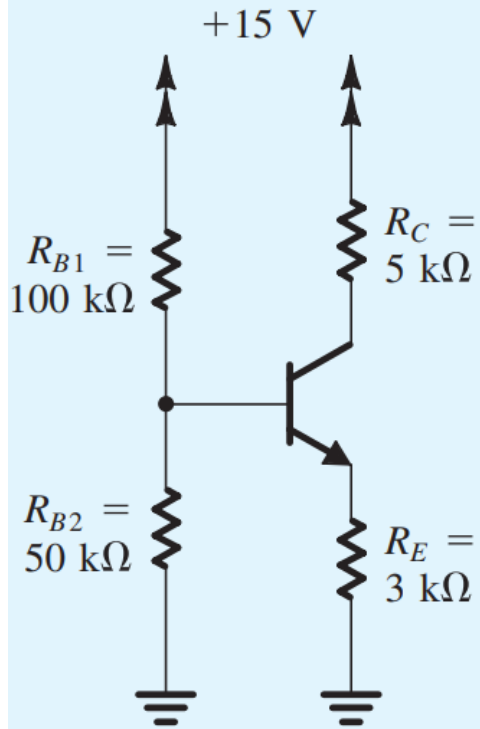
(a)



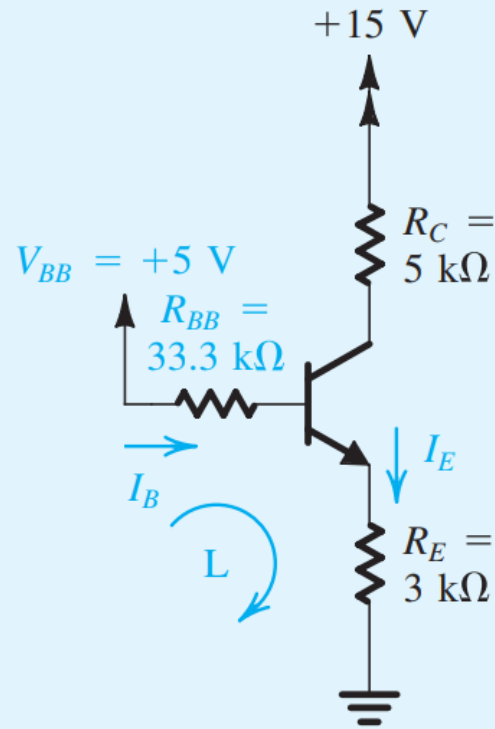
(b)

# BJT Circuits at DC

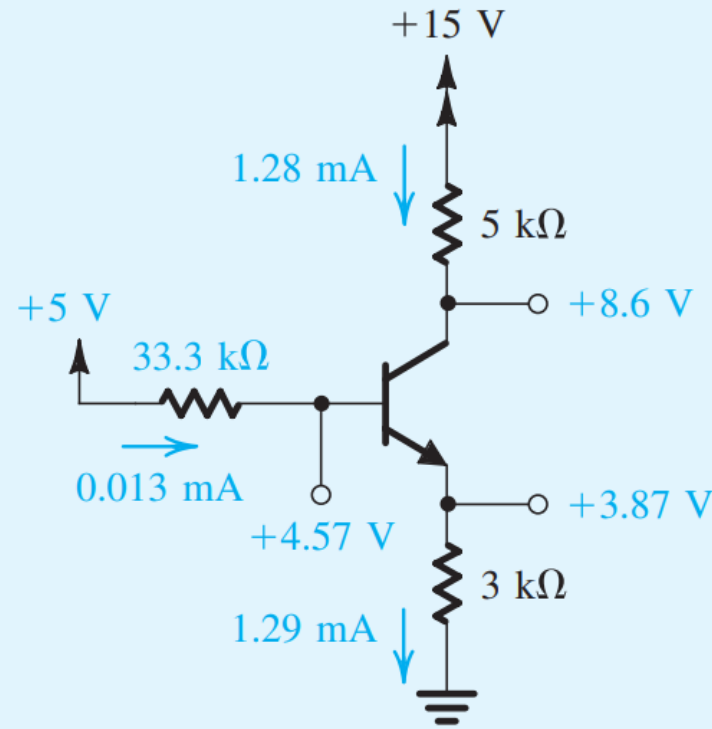
We want to analyze the circuit of Fig. 6.29(a) to determine the voltages at all nodes and the currents through all branches. Assume  $\beta = 100$ .



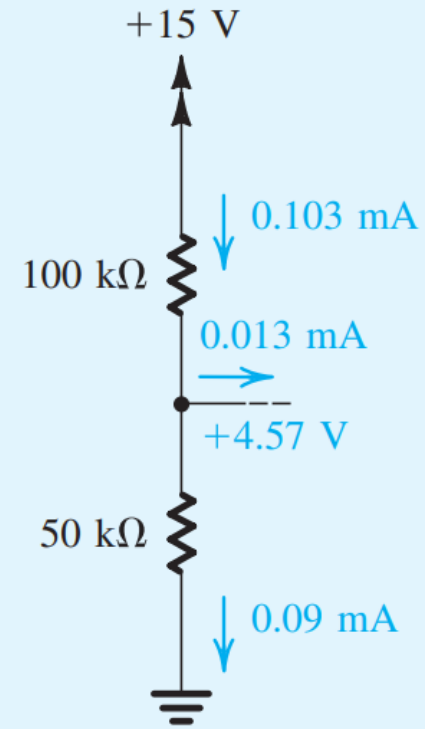
(a)



(b)



(c)

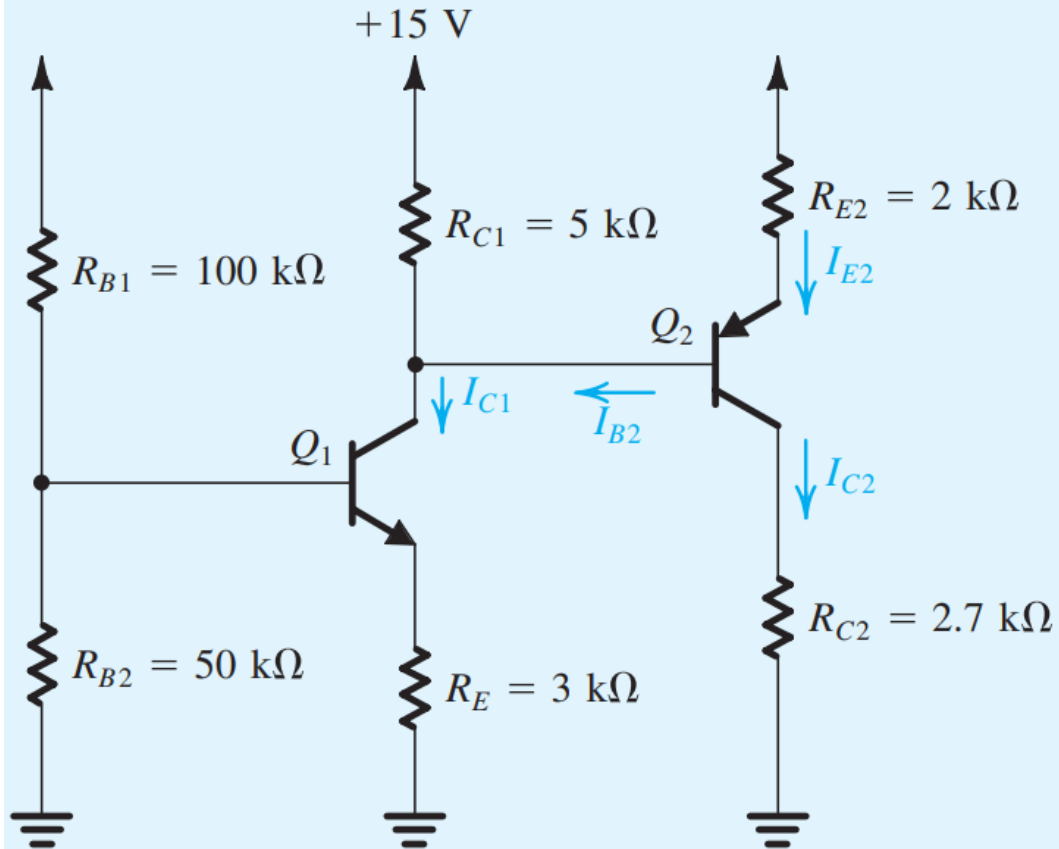


(d)

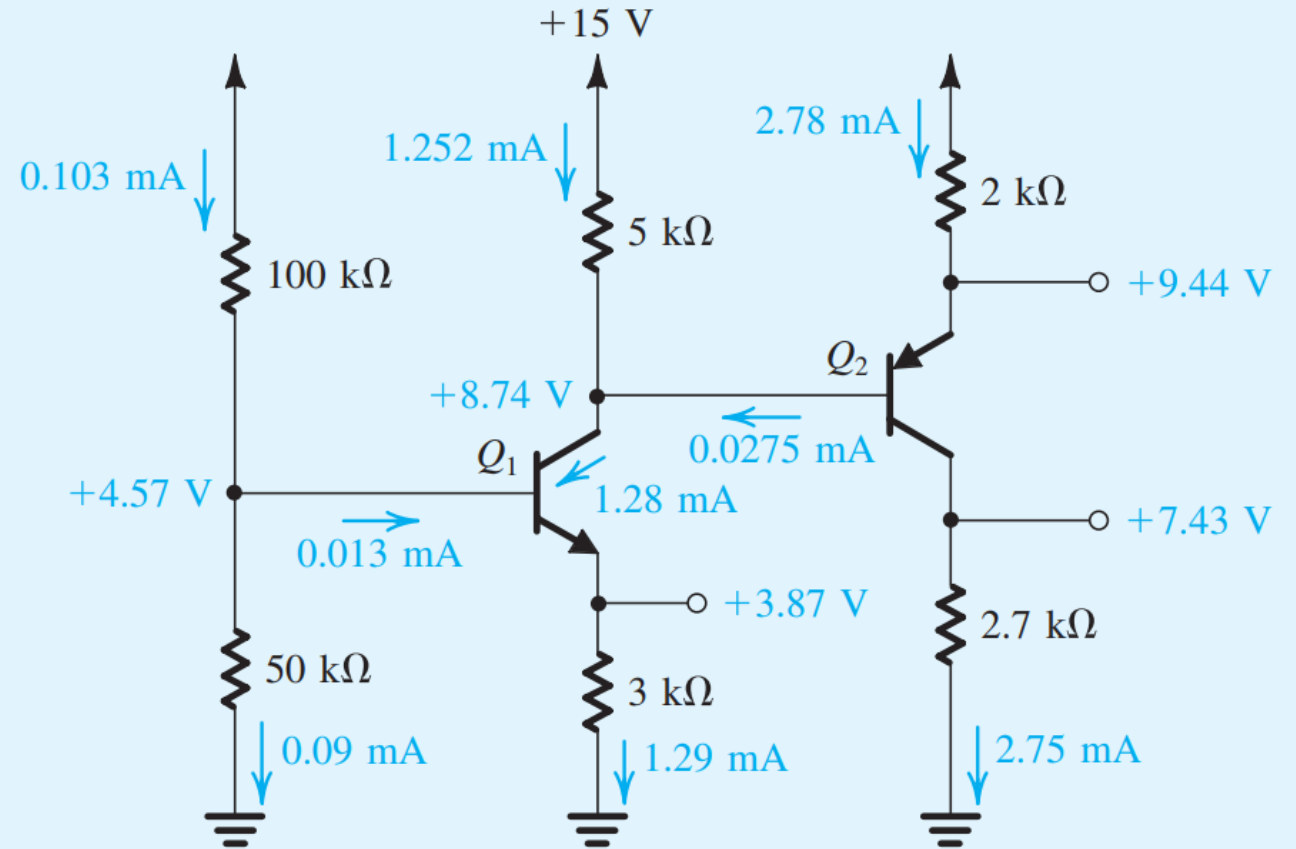


# BJT Circuits at DC

We wish to analyze the circuit in Fig. 6.30(a) to determine the voltages at all nodes and the currents through all branches.



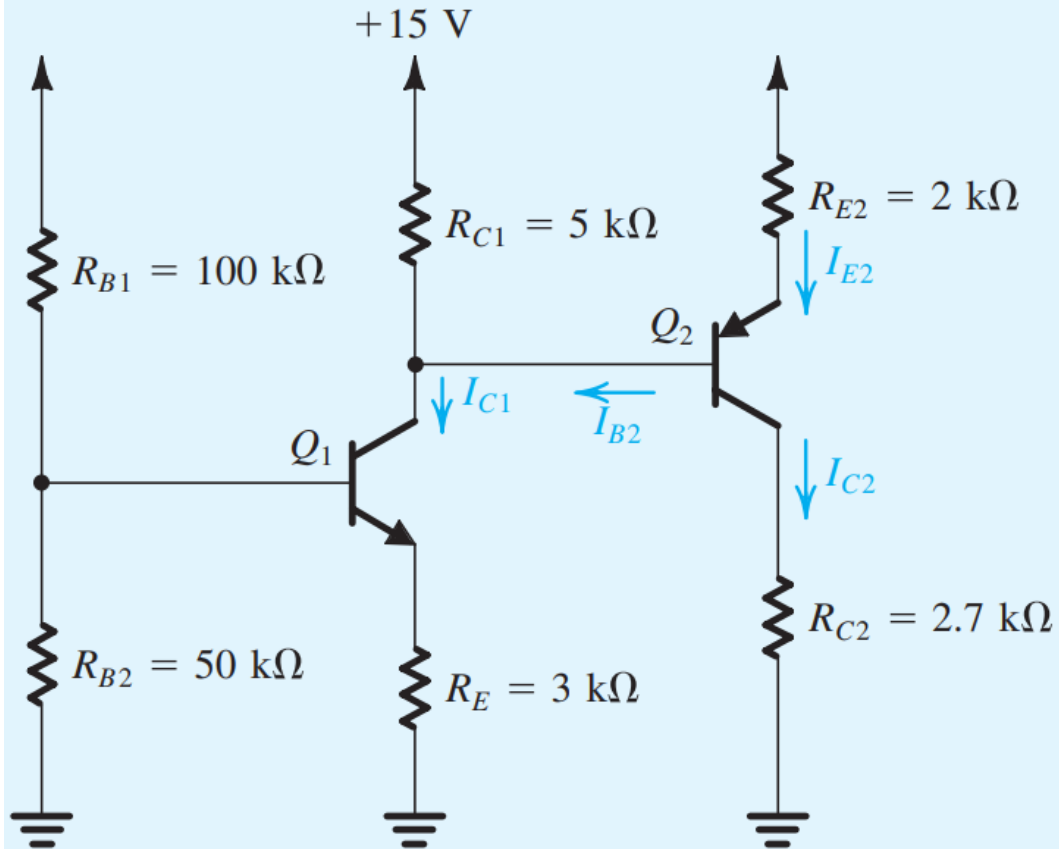
(a)



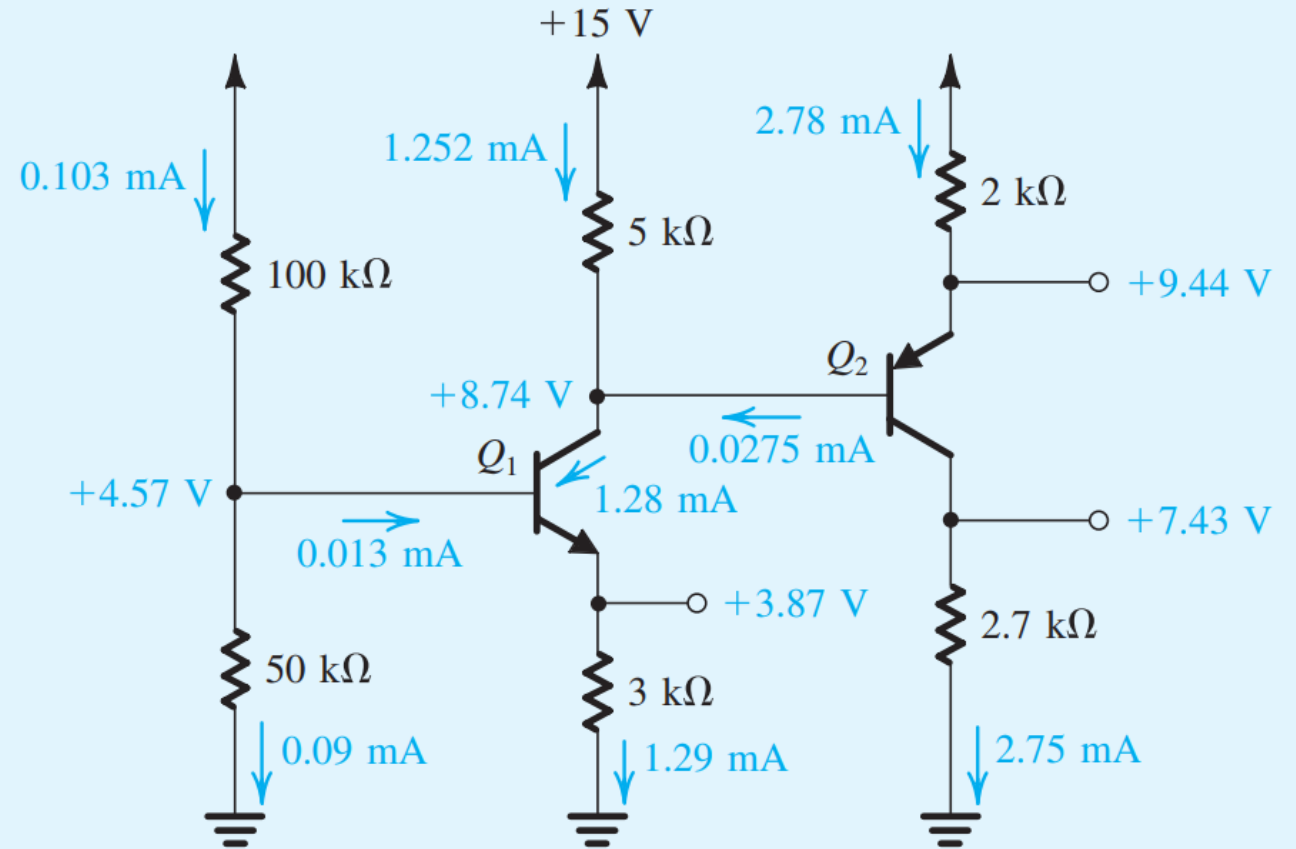
(b)

# BJT Circuits at DC

We wish to analyze the circuit in Fig. 6.30(a) to determine the voltages at all nodes and the currents through all branches.

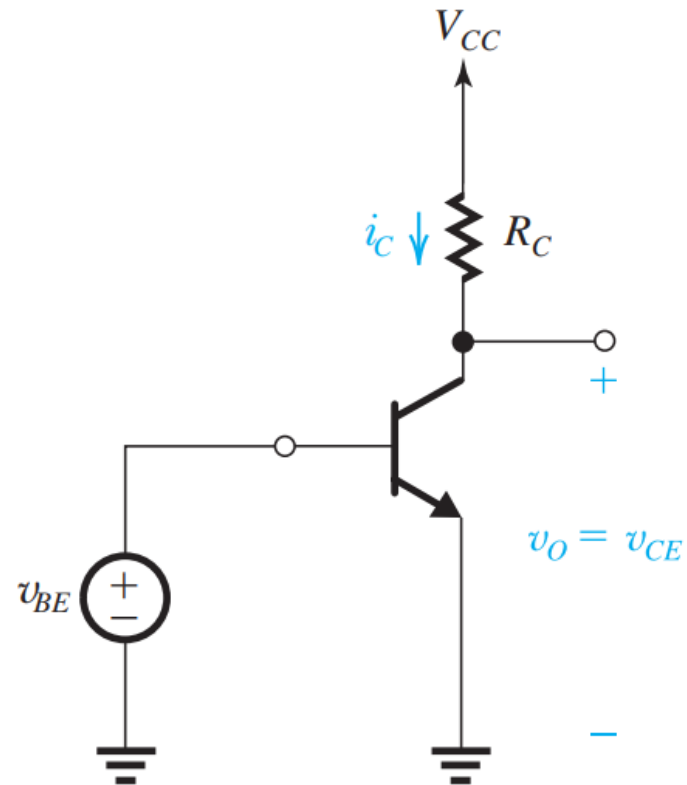


(a)



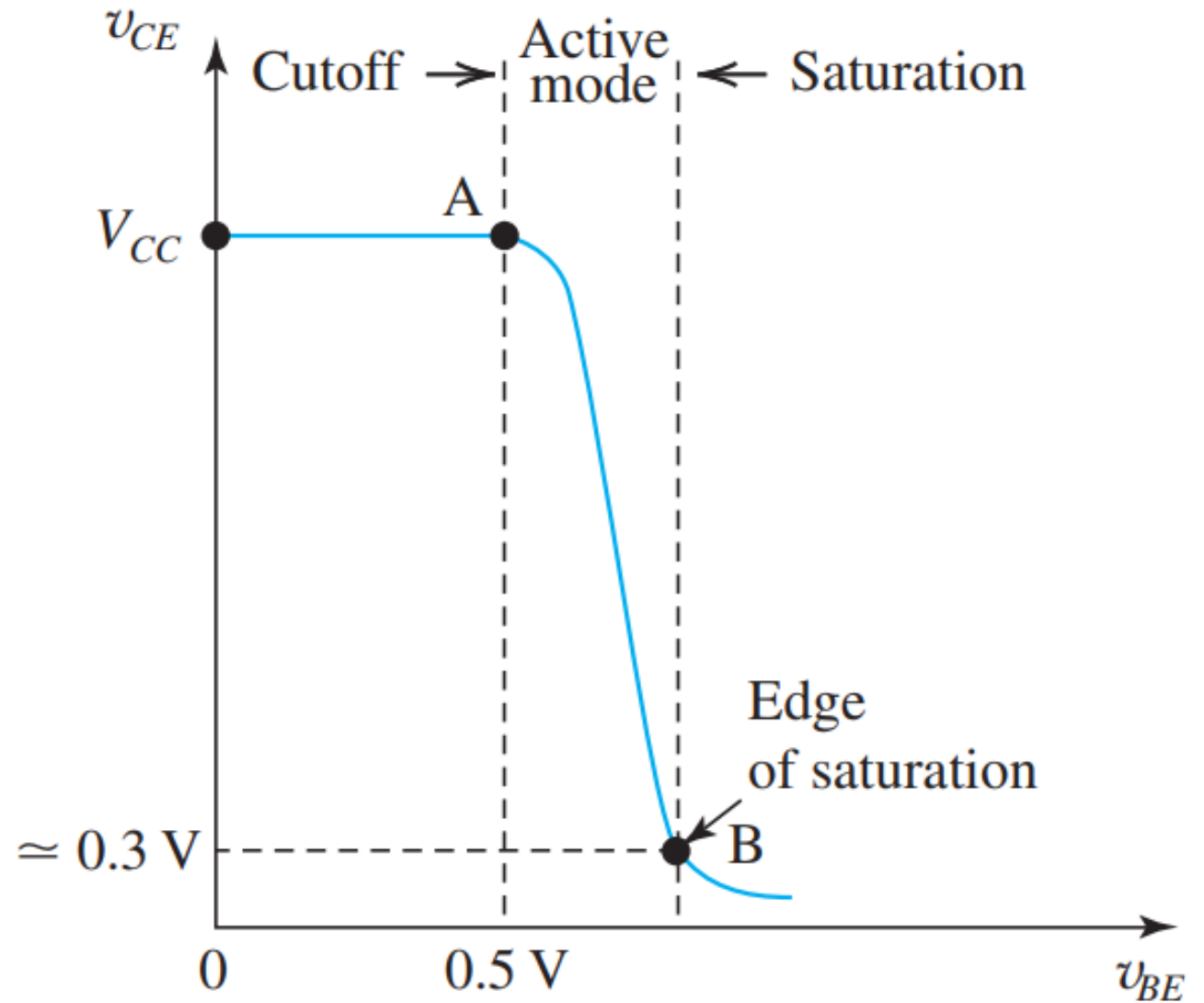
(b)

# Amplifiers

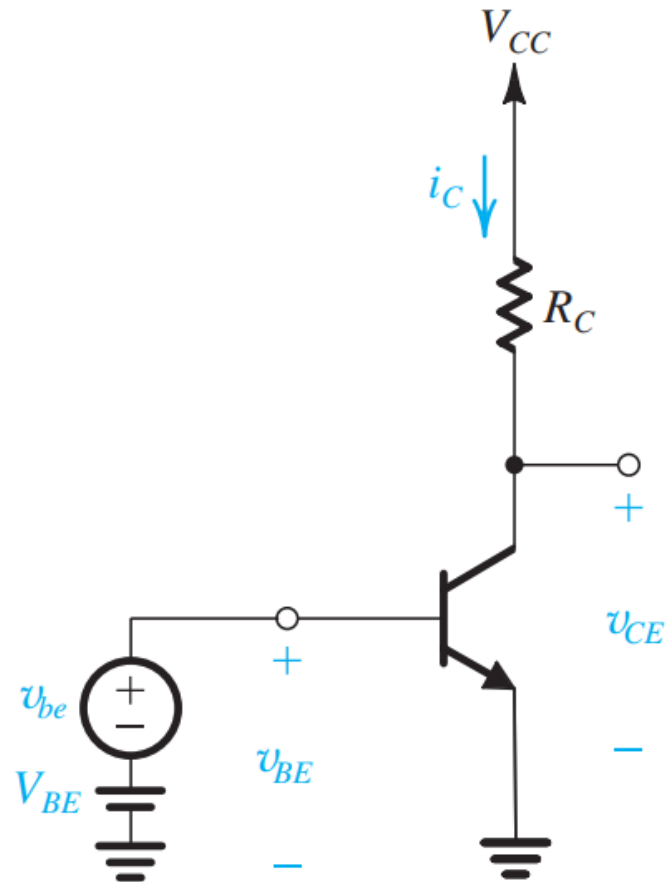


$$v_{CE} = V_{CC} - i_C R_C$$

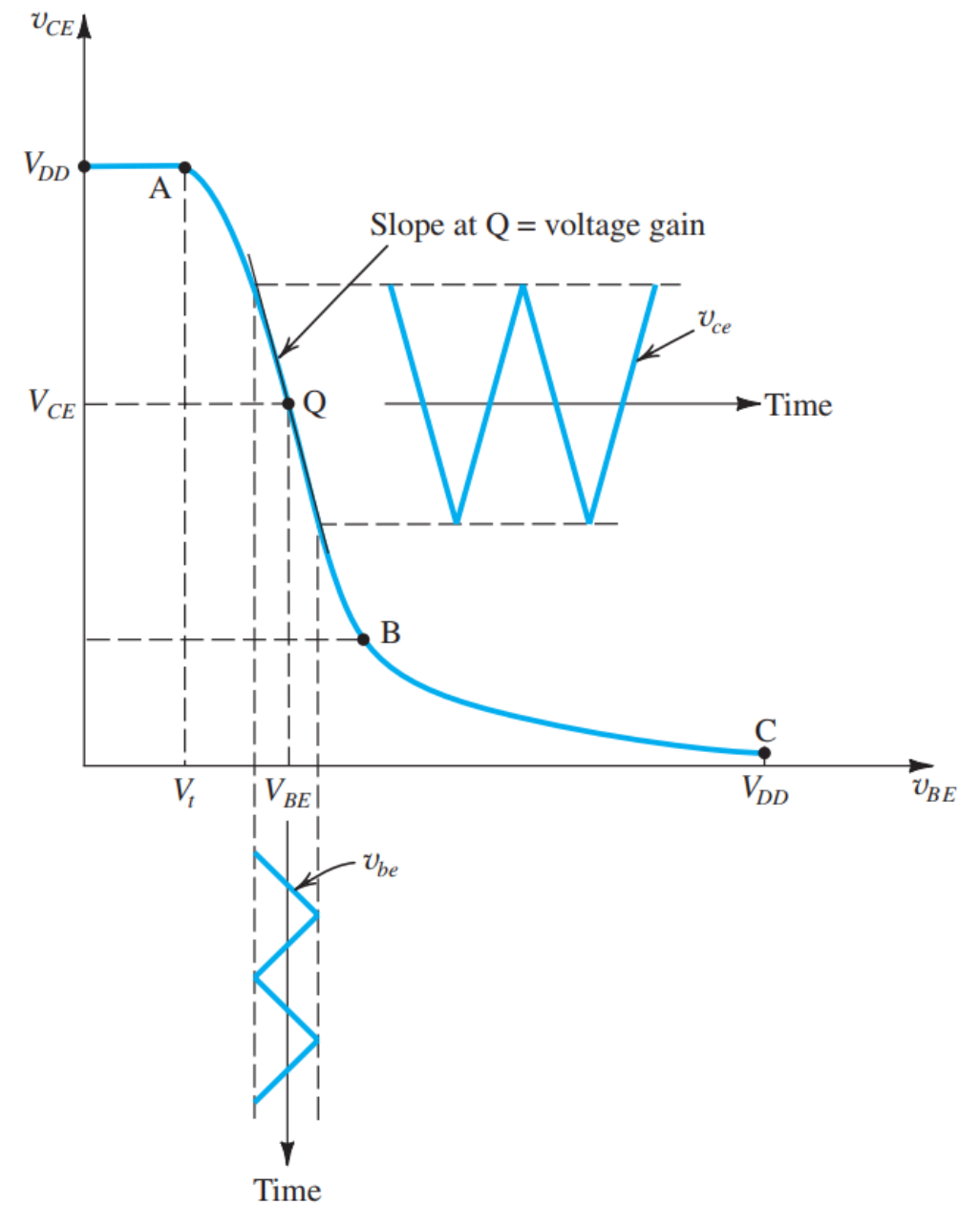
$$v_{CE} = V_{CC} - R_C I_S e^{v_{BE}/V_T}$$



# Amplifiers



$$v_{BE}(t) = V_{BE} + v_{be}(t)$$



# Small Signal Voltage Gain

$$A_v = \left. \frac{dv_{CE}}{dv_{BE}} \right|_{v_{BE}=V_{BE}}$$

$$v_{CE} = V_{CC} - R_C I_S e^{v_{BE}/V_T}$$

$$A_v = -\left(\frac{I_C}{V_T}\right)R_C$$

$$A_v = -\frac{I_C R_C}{V_T}$$

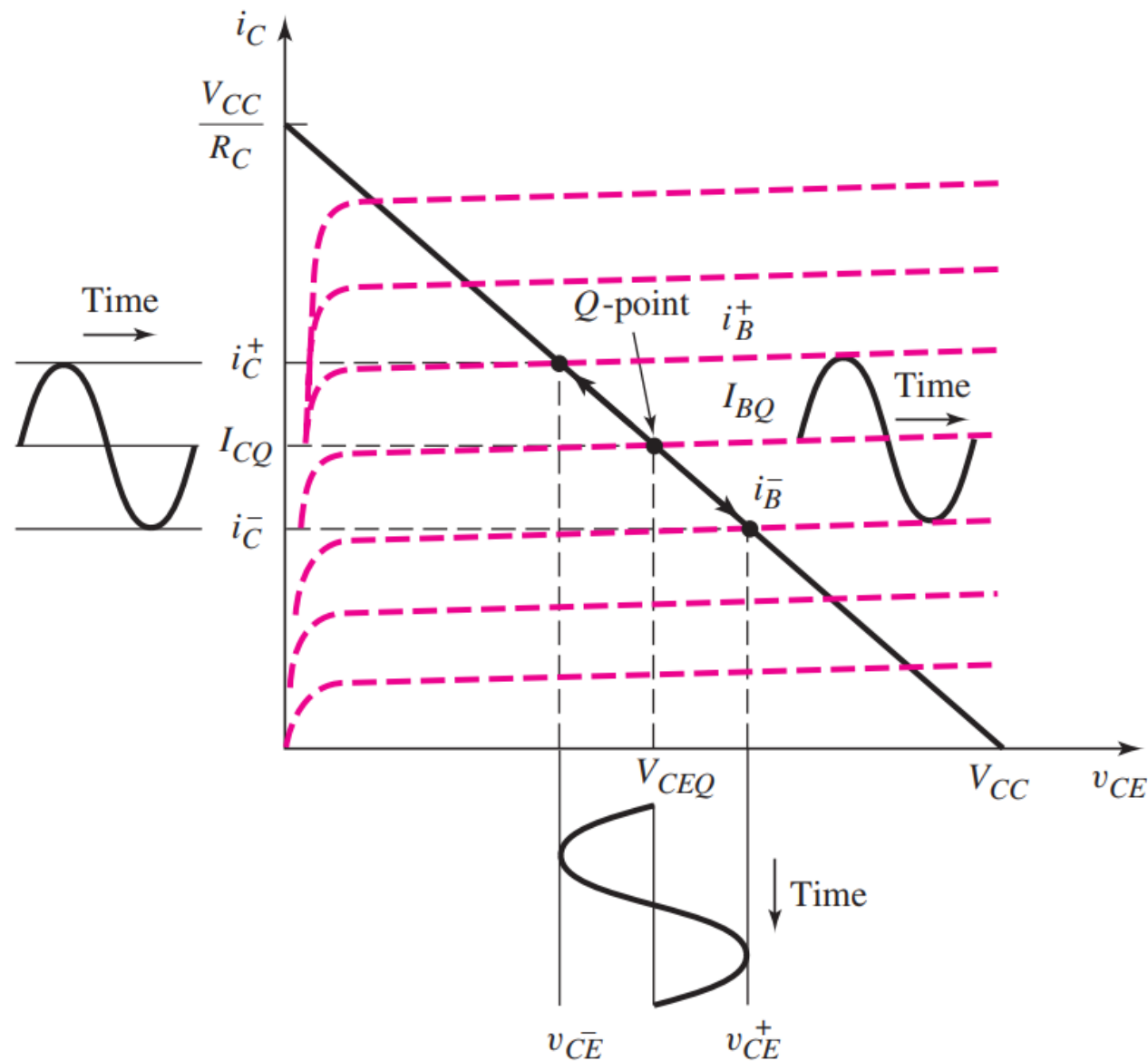
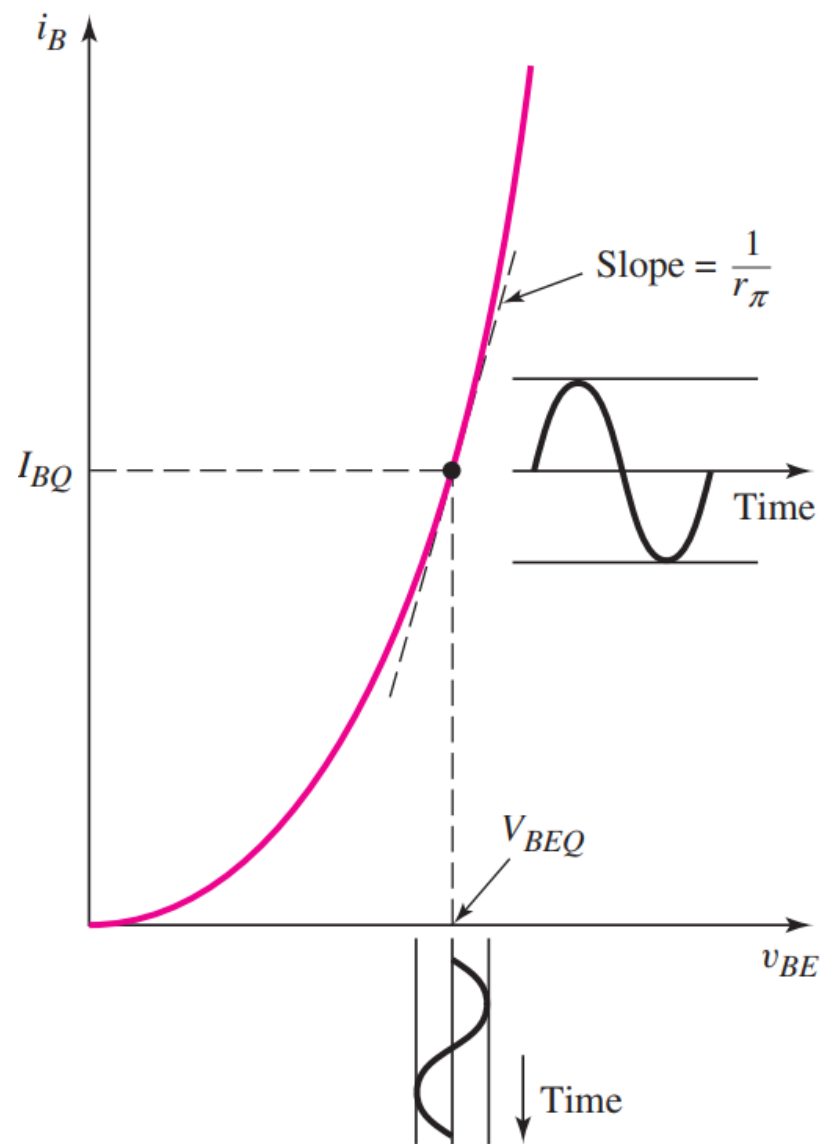
$$A_v = -\frac{V_{CC} - V_{CE}}{V_T}$$

$$|A_{v\max}| = \frac{V_{CC} - 0.3}{V_T}$$

We make the following observations on this expression for the voltage gain:

1. The gain is negative, which signifies that the amplifier is inverting; that is, there is a  $180^\circ$  phase shift between the input and the output. This inversion should have been anticipated from Eq. (7.9).
2. The gain is proportional to the collector bias current  $I_C$  and to the load resistance  $R_C$ .

# Graphical Analysis



# Small Signal Voltage Gain

$i_B, v_{BE}$

Total instantaneous values

$I_B, V_{BE}$

DC values

$i_b, v_{be}$

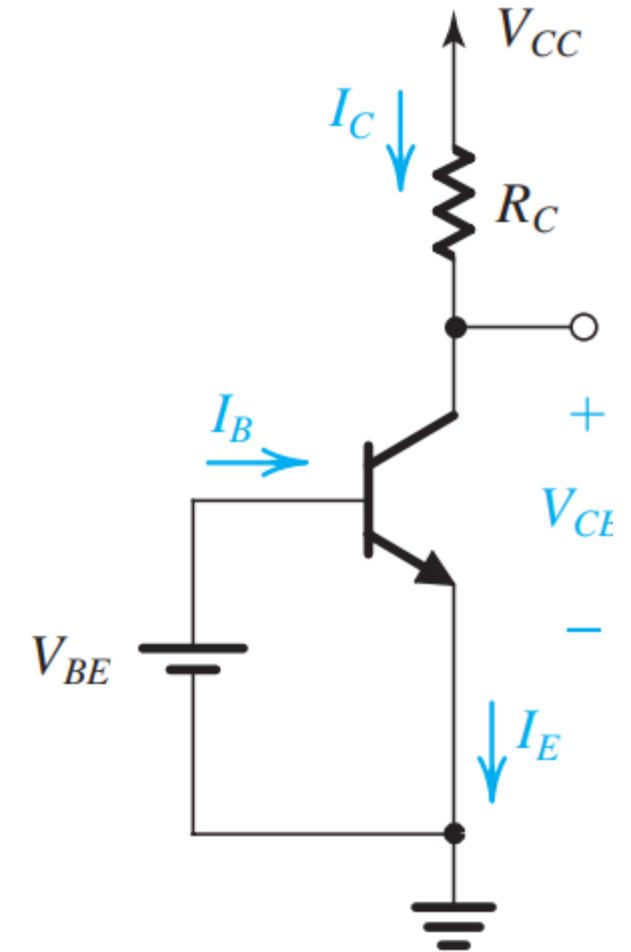
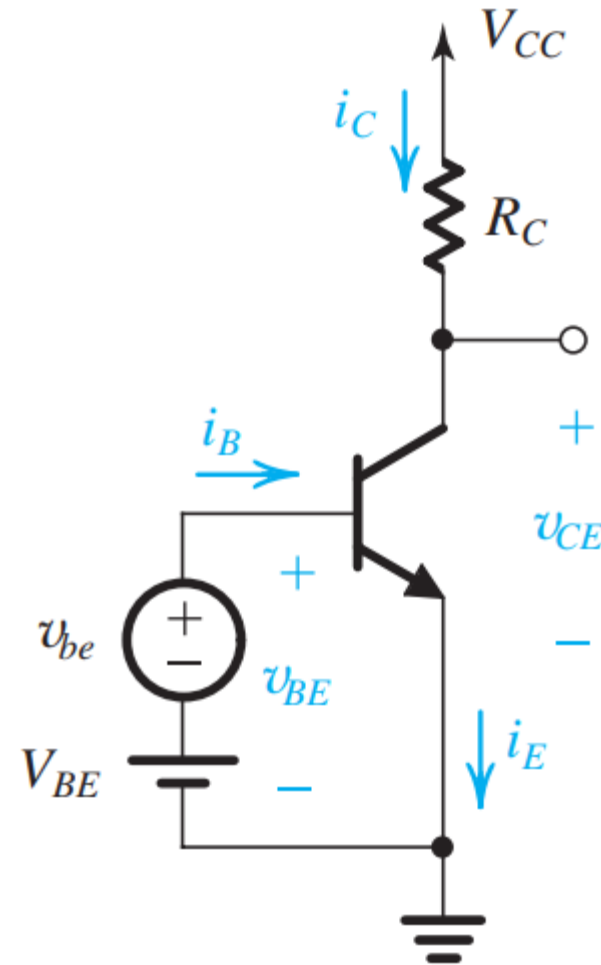
Instantaneous ac values

$$I_C = I_S e^{V_{BE}/V_T}$$

$$I_E = I_C / \alpha$$

$$I_B = I_C / \beta$$

$$V_{CE} = V_{CC} - I_C R_C$$



# Small Signal Voltage Gain

$$v_{BE} = V_{BE} + v_{be}$$

$$i_C = I_S e^{v_{BE}/V_T} = I_S e^{(V_{BE} + v_{be})/V_T} \\ = I_S e^{V_{BE}/V_T} e^{v_{be}/V_T}$$

$$i_C = I_C e^{v_{be}/V_T}$$

$$i_C \simeq I_C \left( 1 + \frac{v_{be}}{V_T} \right) \quad \text{if } v_{be} \ll V_T$$

$$i_C = I_C + \frac{I_C}{V_T} v_{be}$$

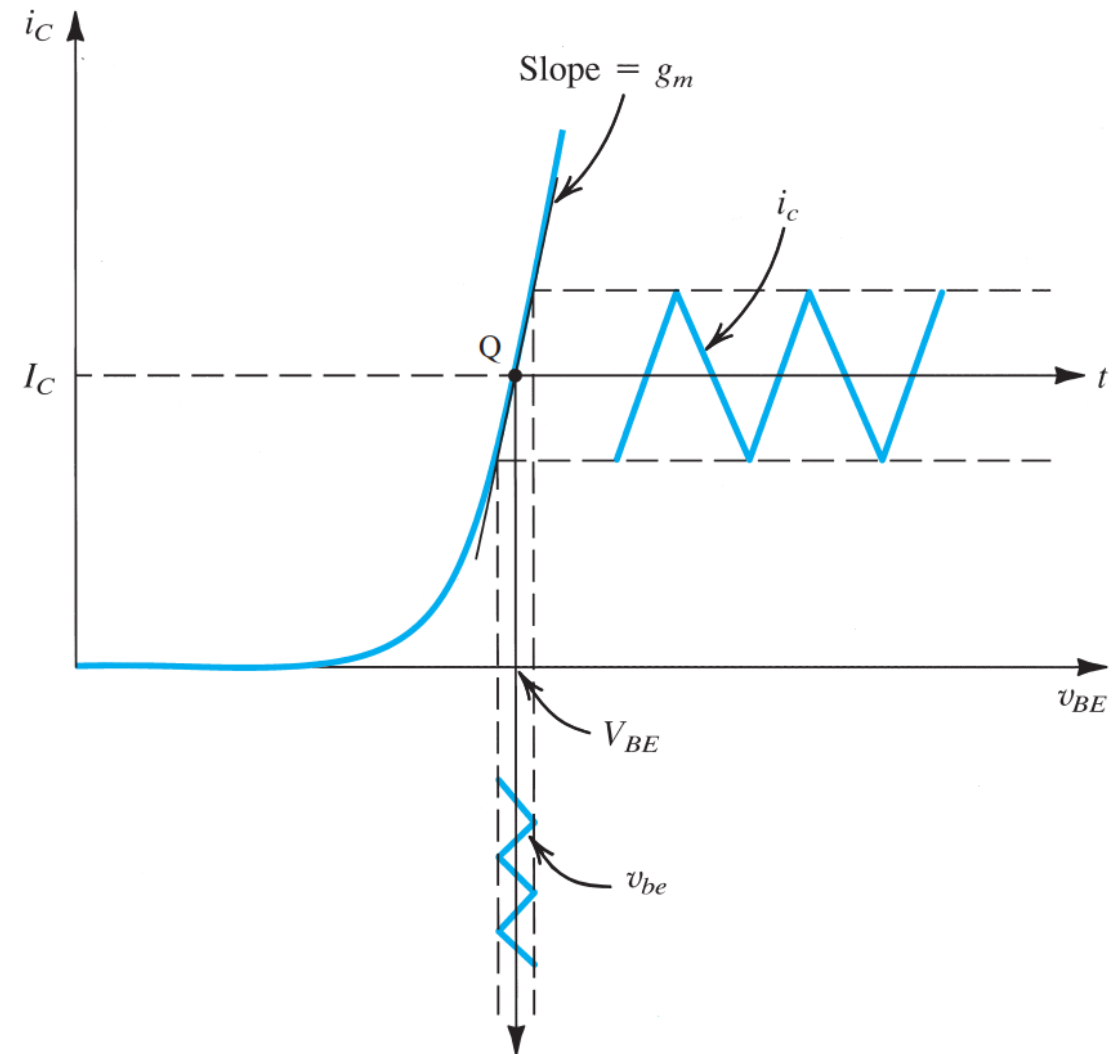
$$i_c = \frac{I_C}{V_T} v_{be}$$

$$i_c = g_m v_{be}$$

$$g_m = \frac{I_C}{V_T}$$

$$g_m = \left. \frac{\partial i_C}{\partial v_{BE}} \right|_{i_C = I_C}$$

where  $g_m$  is the **transconductance**





# Small Signal Voltage Gain

$$i_B = \frac{i_C}{\beta} = \frac{I_C}{\beta} + \frac{1}{\beta} \frac{I_C}{V_T} v_{be}$$

$$i_B = I_B + i_b$$

$$i_b = \frac{1}{\beta} \frac{I_C}{V_T} v_{be}$$

Substituting for  $I_C/V_T$  by  $g_m$  gives

$$i_b = \frac{g_m}{\beta} v_{be}$$

The small-signal input resistance between base and emitter, *looking into the base*, is denoted by  $r_\pi$  and is defined as

$$r_\pi \equiv \frac{v_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$

$$i_E = \frac{i_C}{\alpha} = \frac{I_C}{\alpha} + \frac{i_c}{\alpha}$$

$$i_E = I_E + i_e$$

$$i_e = \frac{i_c}{\alpha} = \frac{I_C}{\alpha V_T} v_{be} = \frac{I_E}{V_T} v_{be}$$

If we denote the small-signal resistance between base and emitter looking into the emitter by  $r_e$ , it can be defined as

$$r_e \equiv \frac{v_{be}}{i_e} = \frac{V_T}{I_E} = \frac{\alpha}{g_m} \simeq \frac{1}{g_m}$$

$$v_{be} = i_b r_\pi = i_e r_e$$

$$r_\pi = (i_e/i_b) r_e = (\beta + 1) r_e$$

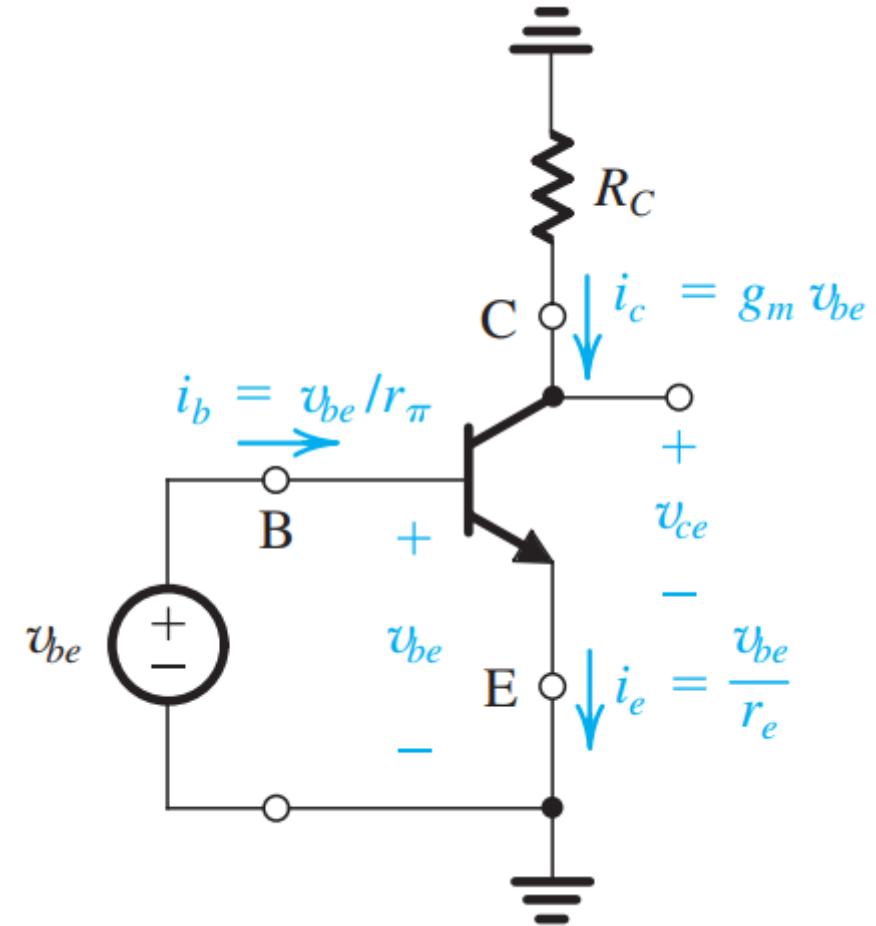
# Small Signal Voltage Gain

$$\begin{aligned}v_{CE} &= V_{CC} - i_C R_C \\&= V_{CC} - (I_C + i_c) R_C \\&= (V_{CC} - I_C R_C) - i_c R_C \\&= V_{CE} - i_c R_C\end{aligned}$$

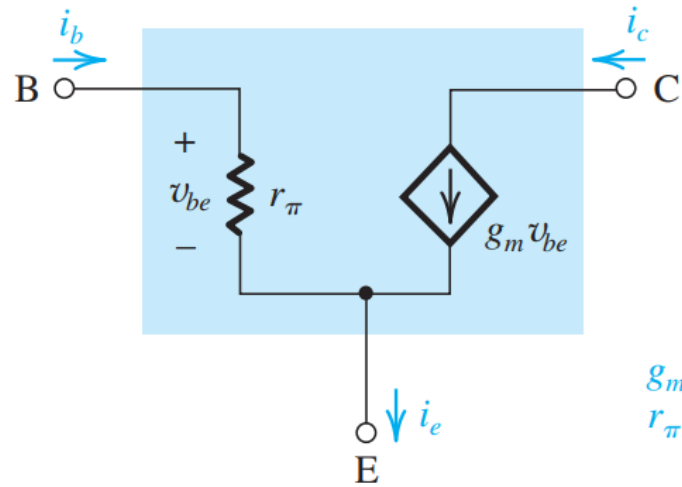
$$\begin{aligned}v_{ce} &= -i_c R_C = -g_m v_{be} R_C \\&= (-g_m R_C) v_{be}\end{aligned}$$

$$A_v \equiv \frac{v_{ce}}{v_{be}} = -g_m R_C$$

$$A_v = -\frac{I_C R_C}{V_T}$$



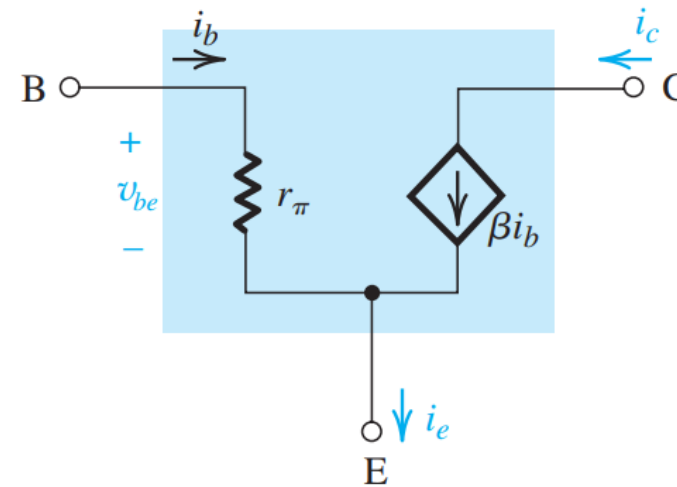
# Small Signal Models (Hybrid- $\pi$ Model)



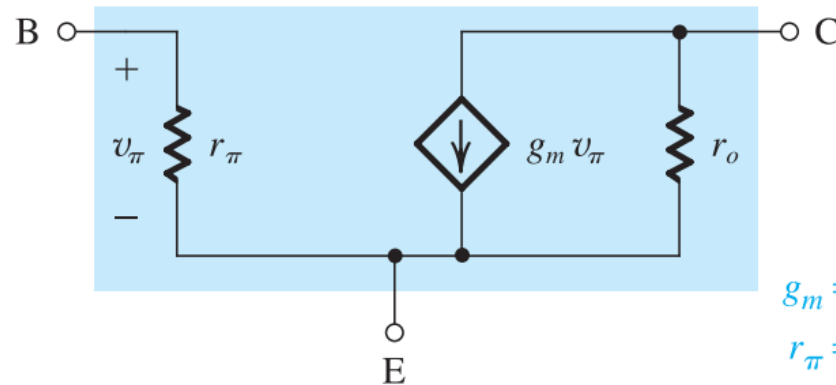
(a)

$$g_m = I_C / V_T$$

$$r_\pi = V_T / I_B = \beta / g_m$$



(b)

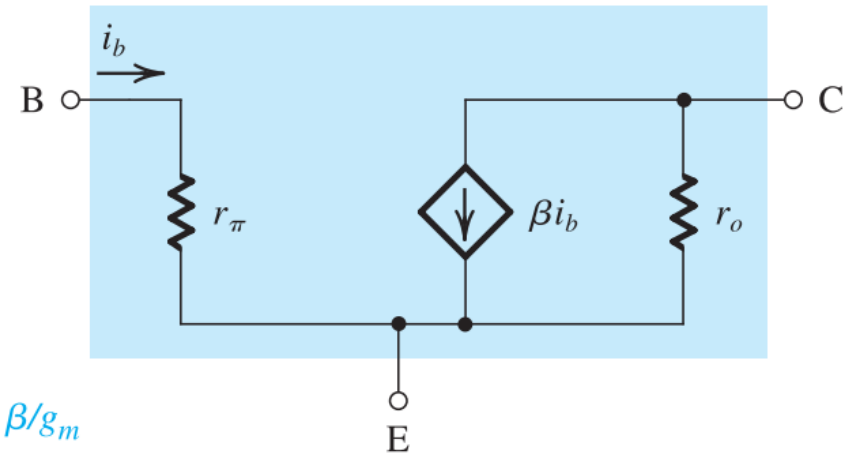


(a)

$$g_m = I_C / V_T$$

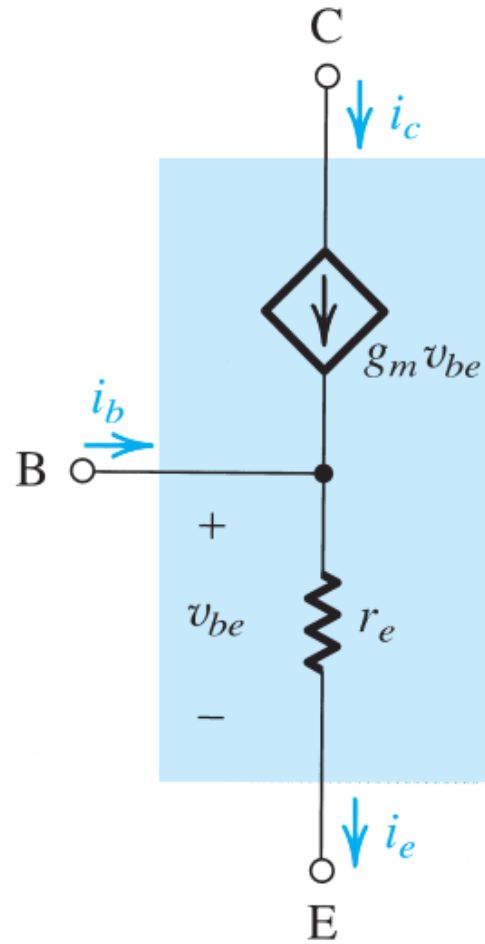
$$r_\pi = V_T / I_B = \beta / g_m$$

$$r_o = V_A / I_C$$



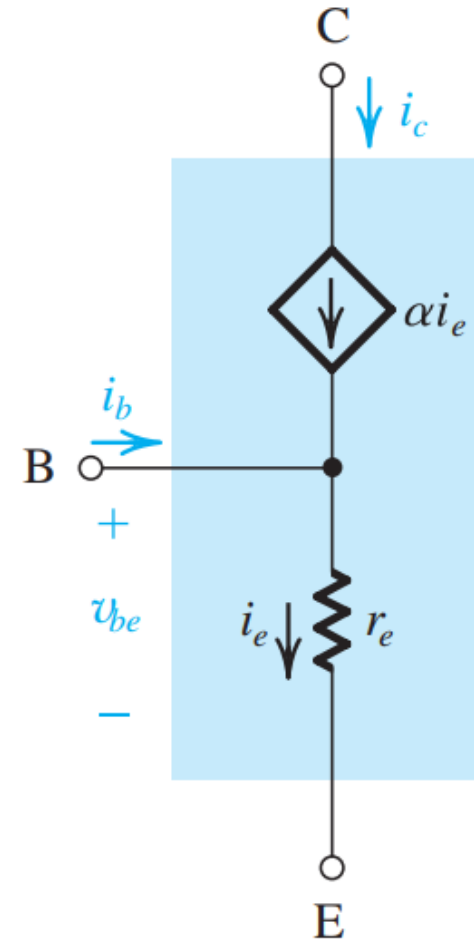
(b)

# Small Signal Models (T Model)



(a)

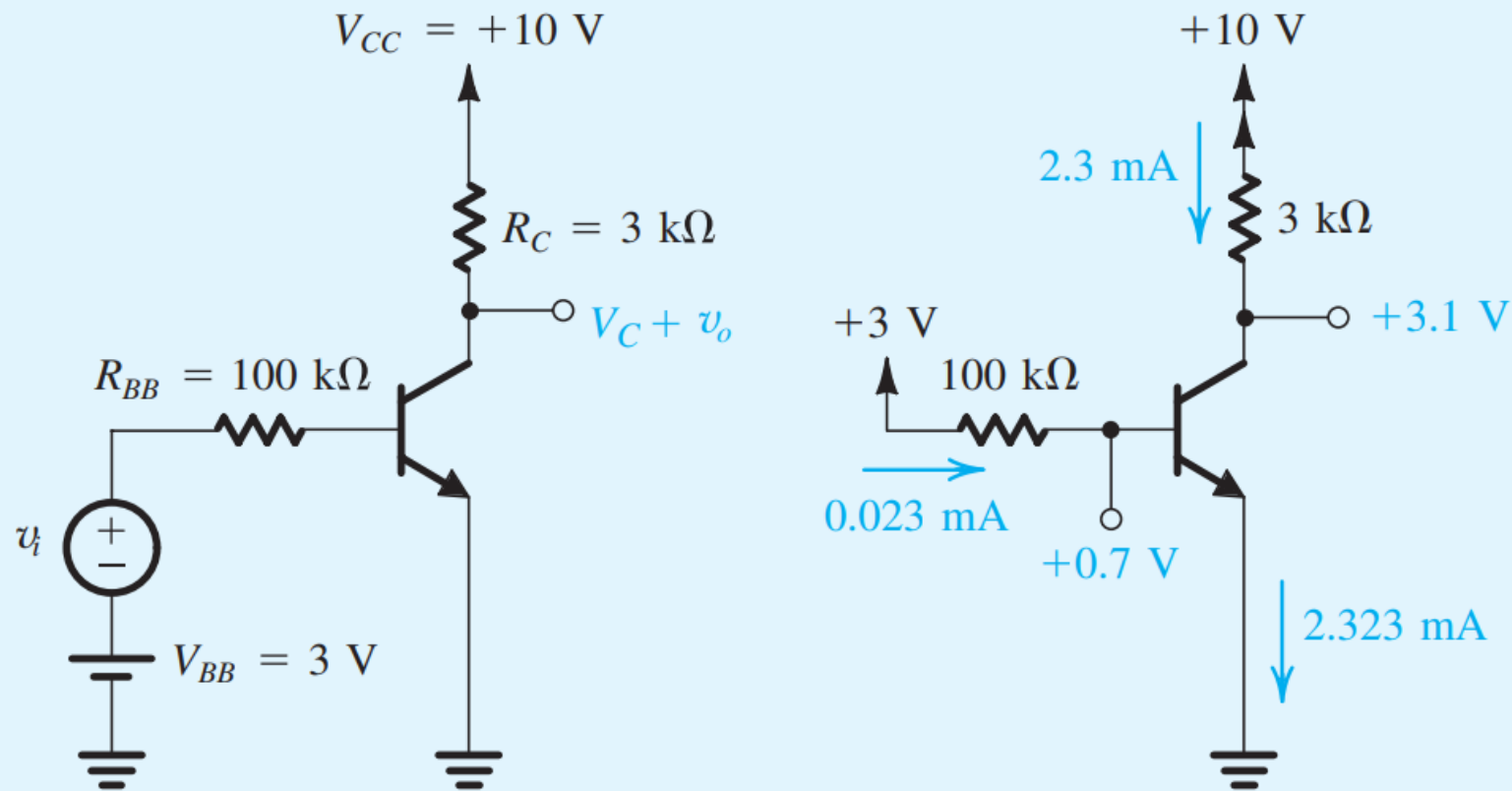
$$g_m = I_C / V_T$$
$$r_e = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$$



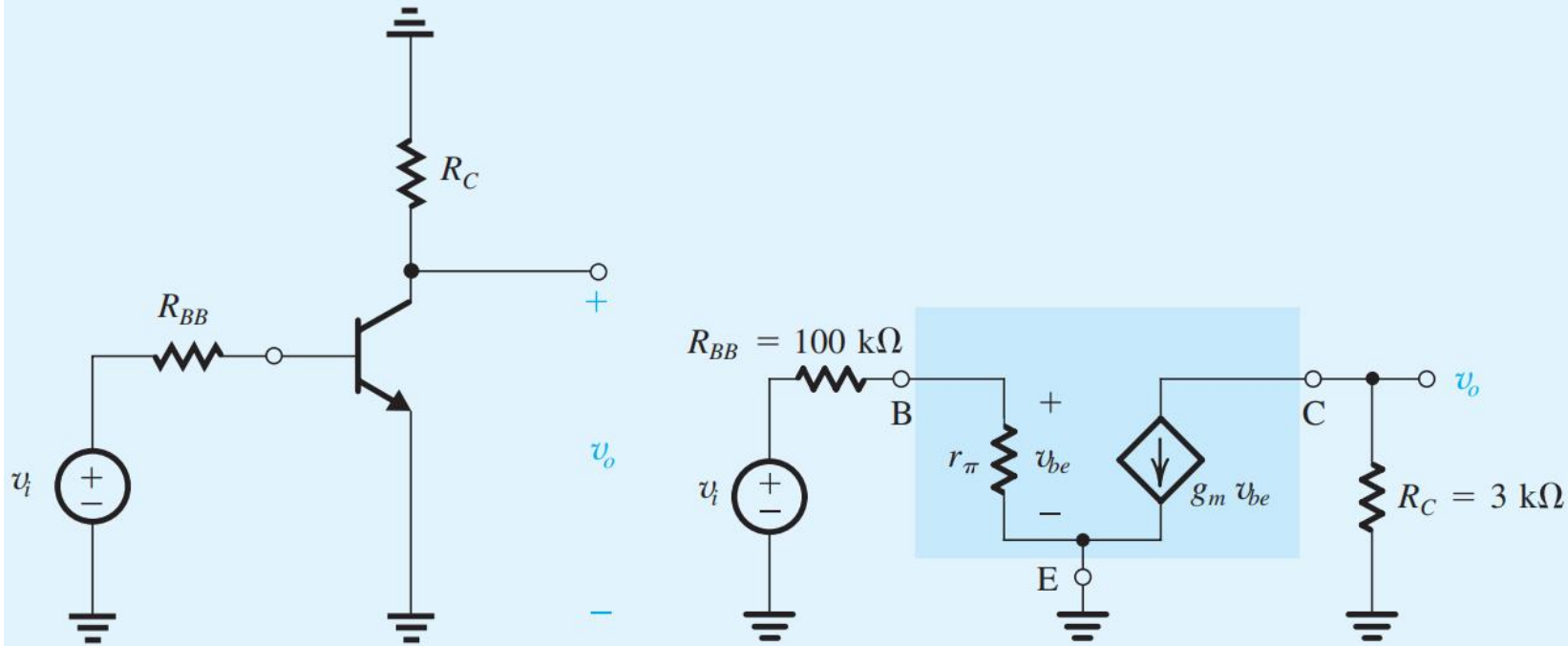
(b)

# Example

We wish to analyze the transistor amplifier shown in Fig. 7.28(a) to determine its voltage gain  $v_o/v_i$ . Assume  $\beta = 100$  and neglect the Early effect.



# Example



$$r_e = \frac{V_T}{I_E} = \frac{25 \text{ mV}}{(2.3/0.99) \text{ mA}} = 10.8 \Omega$$

$$g_m = \frac{I_C}{V_T} = \frac{2.3 \text{ mA}}{25 \text{ mV}} = 92 \text{ mA/V}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{92} = 1.09 \text{ k}\Omega$$

$$v_{be} = v_i \frac{r_\pi}{r_\pi + R_{BB}}$$

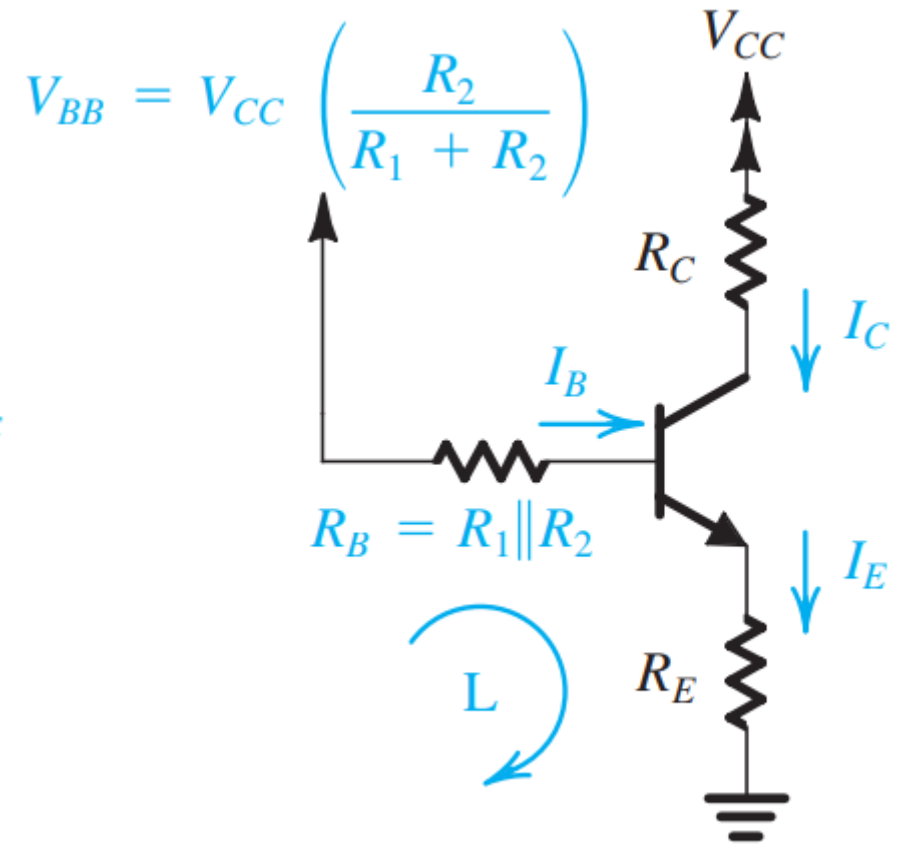
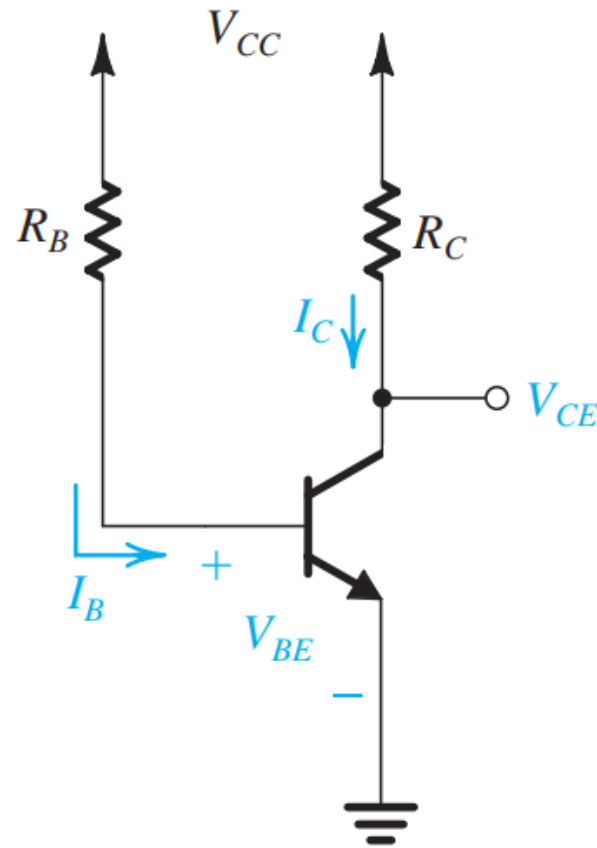
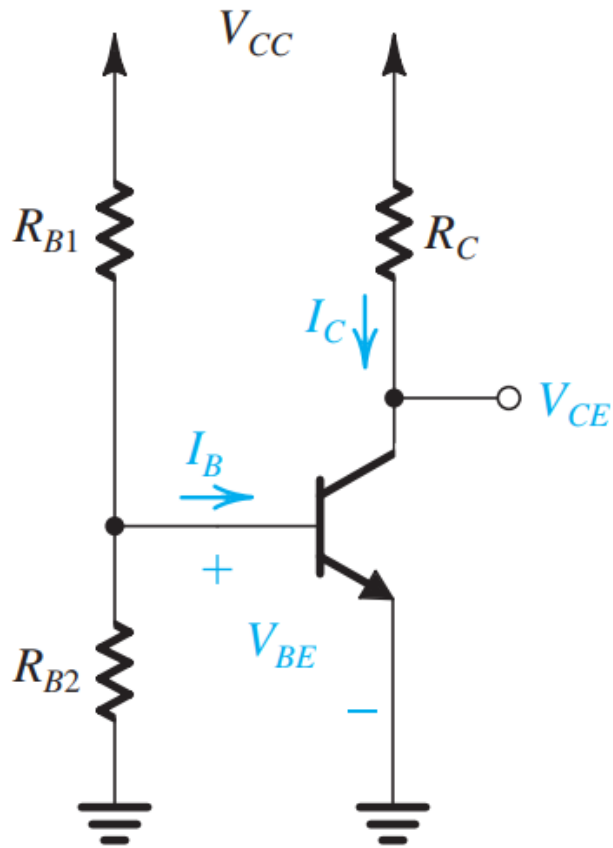
$$= v_i \frac{1.09}{101.09} = 0.011 v_i$$

$$v_o = -g_m v_{be} R_C$$

$$= -92 \times 0.011 v_i \times 3 = -3.04 v_i$$

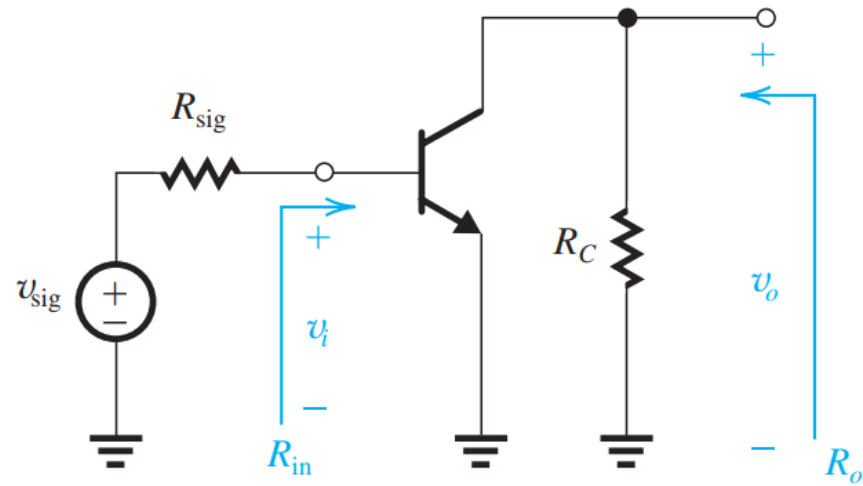
$$A_v = \frac{v_o}{v_i} = -3.04 \text{ V/V}$$

# Biassing Circuit

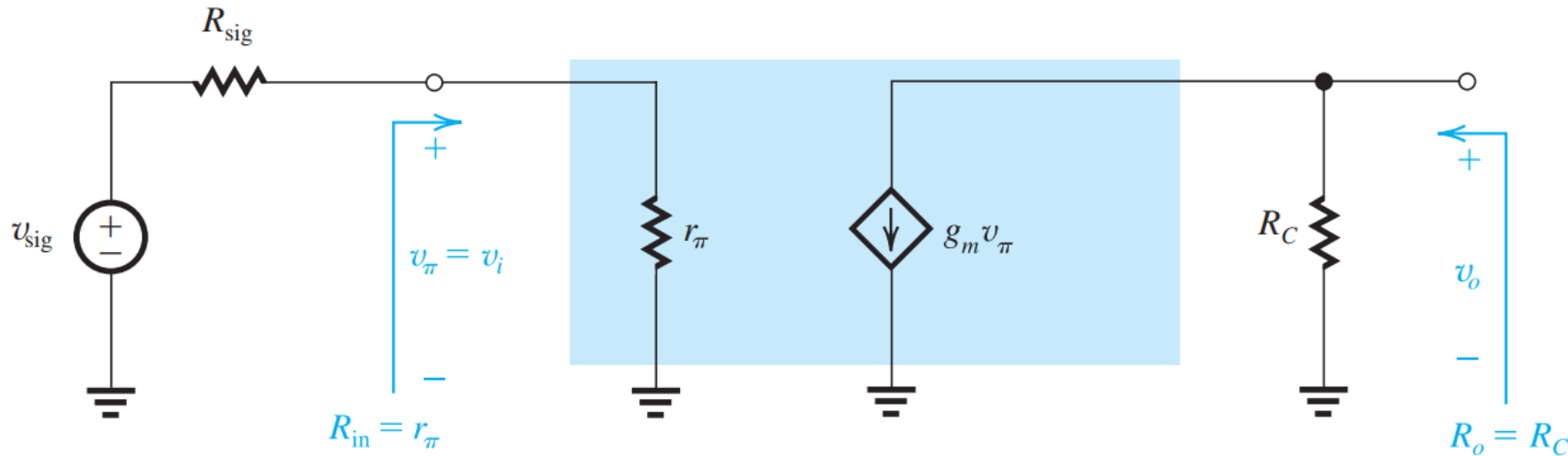


$$I_E = \frac{V_{BB} - V_{BE}}{R_E + R_B/(\beta + 1)}$$

# The Common-Emitter Amplifier (Bias Omitted)



(a)



$$R_{in} = r_{\pi}$$

$$v_i = \frac{r_{\pi}}{r_{\pi} + R_{sig}} v_{sig}$$

$$v_{\pi} = v_i$$

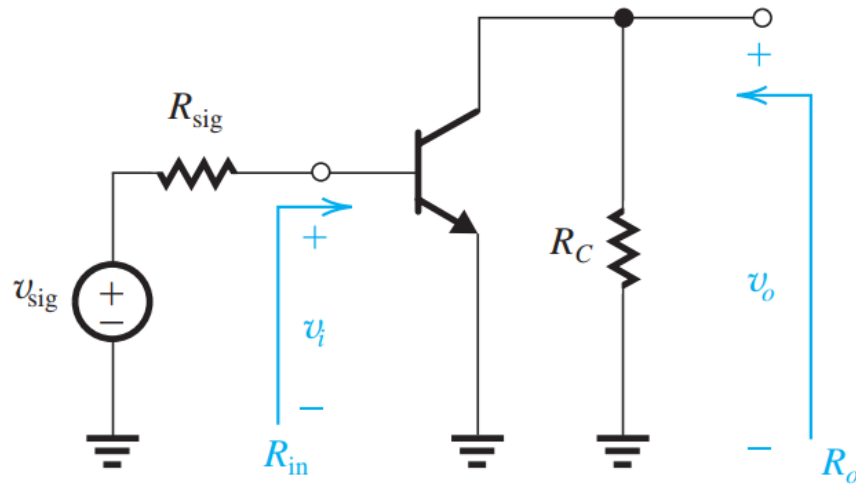
$$v_o = -g_m v_{\pi} R_C$$

$$A_{vo} \equiv \frac{v_o}{v_i} = -g_m R_C$$

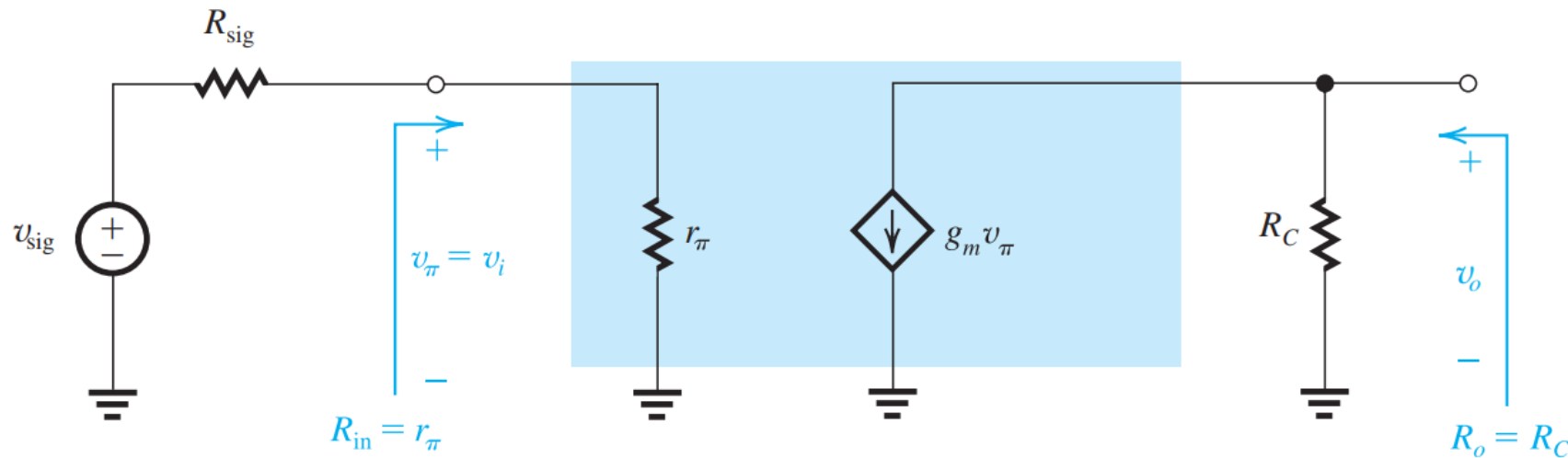
$$R_o = R_C$$



# The Common-Emitter Amplifier (Bias Omitted)



(a)



With a load resistance  $R_L$  connected across  $R_C$ ,

$$A_v = -g_m(R_C \parallel R_L)$$

$$R_o = R_C \parallel R_L$$

$$G_v \equiv \frac{v_o}{v_{\text{sig}}} = \frac{v_i}{v_{\text{sig}}} \frac{v_o}{v_i}$$

$$G_v = -\frac{r_\pi}{r_\pi + R_{\text{sig}}} g_m (R_C \parallel R_L)$$

# Common-Emitter Amplifier

A CE amplifier utilizes a BJT with  $\beta = 100$  is biased at  $I_C = 1 \text{ mA}$  and has a collector resistance  $R_C = 5 \text{ k}\Omega$ . Find  $R_{in}$ ,  $R_o$ , and  $A_{vo}$ . If the amplifier is fed with a signal source having a resistance of  $5 \text{ k}\Omega$ , and a load resistance  $R_L = 5 \text{ k}\Omega$  is connected to the output terminal, find the resulting  $A_v$  and  $G_v$ . If  $\hat{v}_\pi$  is to be limited to  $5 \text{ mV}$ , what are the corresponding  $\hat{v}_{sig}$  and  $\hat{v}_o$  with the load connected?

## Solution

The amplifier characteristic parameters can now be found as

At  $I_C = 1 \text{ mA}$ ,

$$g_m = \frac{I_C}{V_T} = \frac{1 \text{ mA}}{0.025 \text{ V}} = 40 \text{ mA/V}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{40 \text{ mA/V}} = 2.5 \text{ k}\Omega$$

$$R_{in} = r_\pi = 2.5 \text{ k}\Omega$$

$$\begin{aligned} A_{vo} &= -g_m R_C \\ &= -40 \text{ mA/V} \times 5 \text{ k}\Omega \\ &= -200 \text{ V/V} \end{aligned}$$

$$R_o = R_C = 5 \text{ k}\Omega$$

$$\begin{aligned} A_v &= A_{vo} \frac{R_L}{R_L + R_o} \\ &= -200 \times \frac{5}{5 + 5} = -100 \text{ V/V} \end{aligned}$$

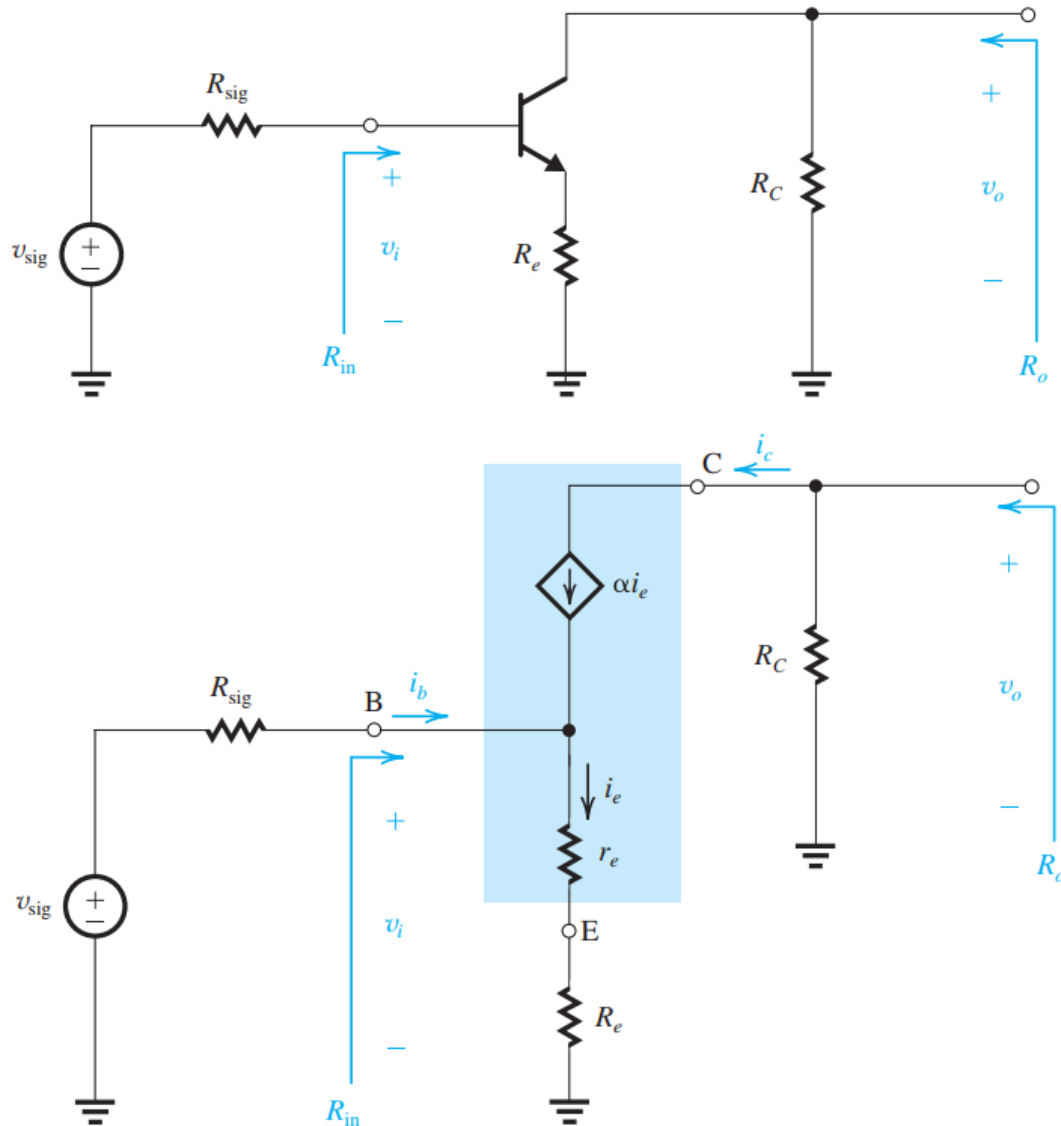
$$\begin{aligned} A_v &= -g_m (R_C \parallel R_L) \\ &= -40(5 \parallel 5) = -100 \text{ V/V} \end{aligned}$$

$$\begin{aligned} G_v &= \frac{R_{in}}{R_{in} + R_{sig}} A_v \\ &= \frac{2.5}{2.5 + 5} \times -100 = -33.3 \text{ V/V} \end{aligned}$$

$$\hat{v}_{sig} = \left( \frac{R_{in} + R_{sig}}{R_{in}} \right) \hat{v}_\pi = \frac{2.5 + 5}{2.5} \times 5 = 15 \text{ mV}$$

$$\hat{v}_o = G_v \hat{v}_{sig} = 33.3 \times 0.015 = 0.5 \text{ V}$$

# The CE Amplifier with Emitter Resistance (Bias Omitted)



$$R_{in} \equiv \frac{v_i}{i_b}$$

$$i_b = (1 - \alpha)i_e = \frac{i_e}{\beta + 1}$$

$$i_e = \frac{v_i}{r_e + R_e}$$

The input resistance looking into the base is  $(\beta + 1)$  times the total resistance in the emitter, and is known as the **resistance-reflection rule**.

$$R_{in} = (\beta + 1)(r_e + R_e)$$

$$\begin{aligned} \frac{R_{in}(\text{with } R_e \text{ included})}{R_{in}(\text{without } R_e)} &= \frac{(\beta + 1)(r_e + R_e)}{(\beta + 1)r_e} \\ &= 1 + \frac{R_e}{r_e} \simeq 1 + g_m R_e \end{aligned}$$

# The CE Amplifier with Emitter Resistance (Bias Omitted)

$$\begin{aligned}v_o &= -i_c R_C \\ &= -\alpha i_e R_C\end{aligned}$$

$$A_{vo} = -\alpha \frac{R_C}{r_e + R_e}$$

Voltage gain from base to collector

$$= -\alpha \frac{\text{Total resistance in collector}}{\text{Total resistance in emitter}}$$

$$A_{vo} = -\frac{\alpha}{r_e} \frac{R_C}{1 + R_e/r_e}$$

$$R_o = R_C$$

$$\begin{aligned}A_v &= A_{vo} \frac{R_L}{R_L + R_o} \\ &= -\alpha \frac{R_C}{r_e + R_e} \frac{R_L}{R_L + R_C} \\ &= -\alpha \frac{R_C \parallel R_L}{r_e + R_e}\end{aligned}$$

$$G_v = -\beta \frac{R_C \parallel R_L}{R_{\text{sig}} + (\beta + 1)(r_e + R_e)}$$

$$G_v = -\beta \frac{R_C \parallel R_L}{R_{\text{sig}} + (\beta + 1)r_e}$$

$$\frac{v_\pi}{v_i} = \frac{r_e}{r_e + R_e} \simeq \frac{1}{1 + g_m R_e}$$

# The CE Amplifier with Emitter Resistance (Bias Omitted)

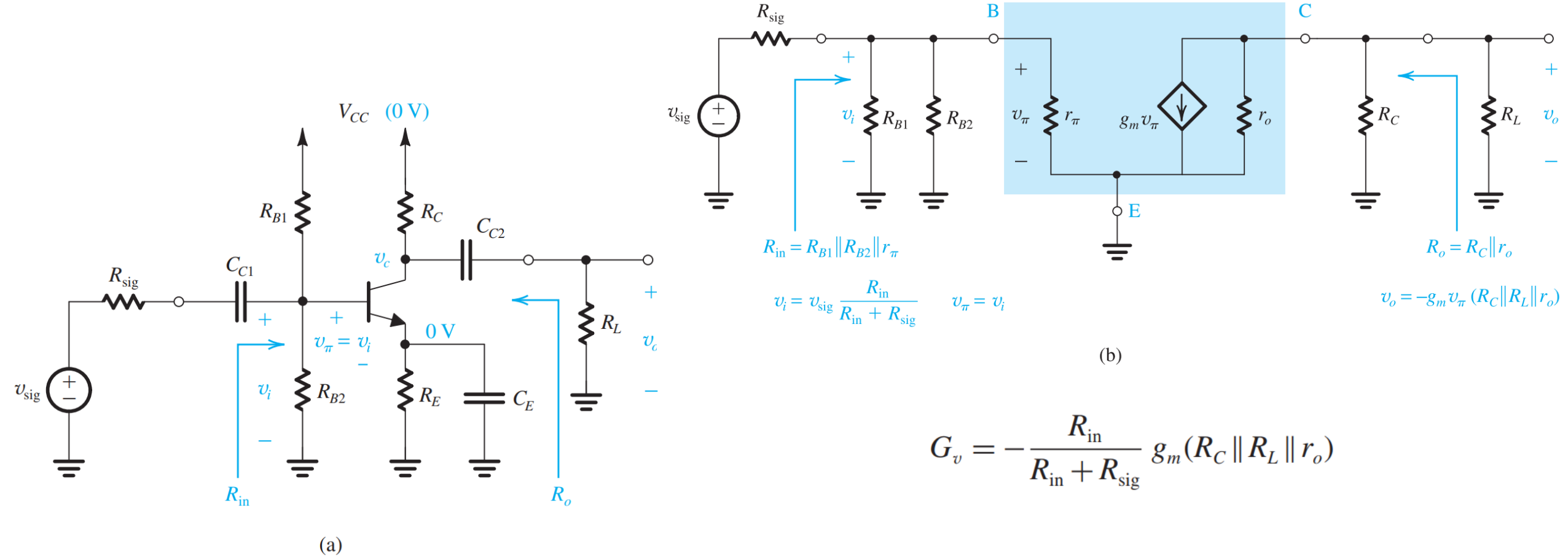
To summarize, including a resistance  $R_e$  in the emitter of the CE amplifier results in the following characteristics:

1. The input resistance  $R_{in}$  is increased by the factor  $(1 + g_m R_e)$ .
2. The voltage gain from base to collector,  $A_v$ , is reduced by the factor  $(1 + g_m R_e)$ .
3. For the same nonlinear distortion, the input signal  $v_i$  can be increased by the factor  $(1 + g_m R_e)$ .
4. The overall voltage gain is less dependent on the value of  $\beta$ .
5. The high-frequency response is significantly improved.

With the exception of gain reduction, these characteristics represent performance improvements. Indeed, the reduction in gain is the price paid for obtaining the other performance improvements. In many cases this is a good bargain; it is the underlying philosophy for the use of negative feedback. Finally, we note that the negative-feedback action of  $R_e$  gives it the name **emitter degeneration resistance**.

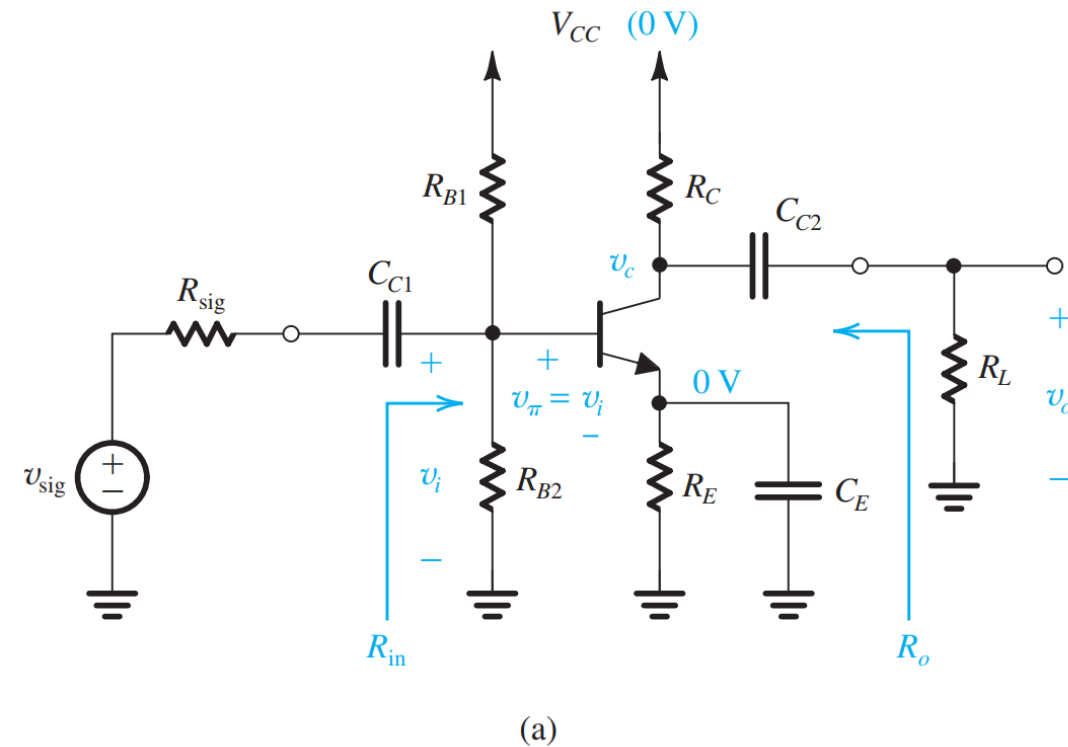


# The CE Amplifier with Emitter Resistance (With Bias)



**Figure 7.56** (a) A common-emitter amplifier using the classical biasing arrangement of Fig. 7.52(a).  
(b) Equivalent circuit and analysis.

# Types of Capacitances



**Input Coupling Capacitor ( $C_{C1}$ ):** Placed between the AC signal source and the transistor's base.

- **Function:** Blocks the DC biasing voltage of the BJT from leaking into the signal source, while allowing the AC signal to pass.

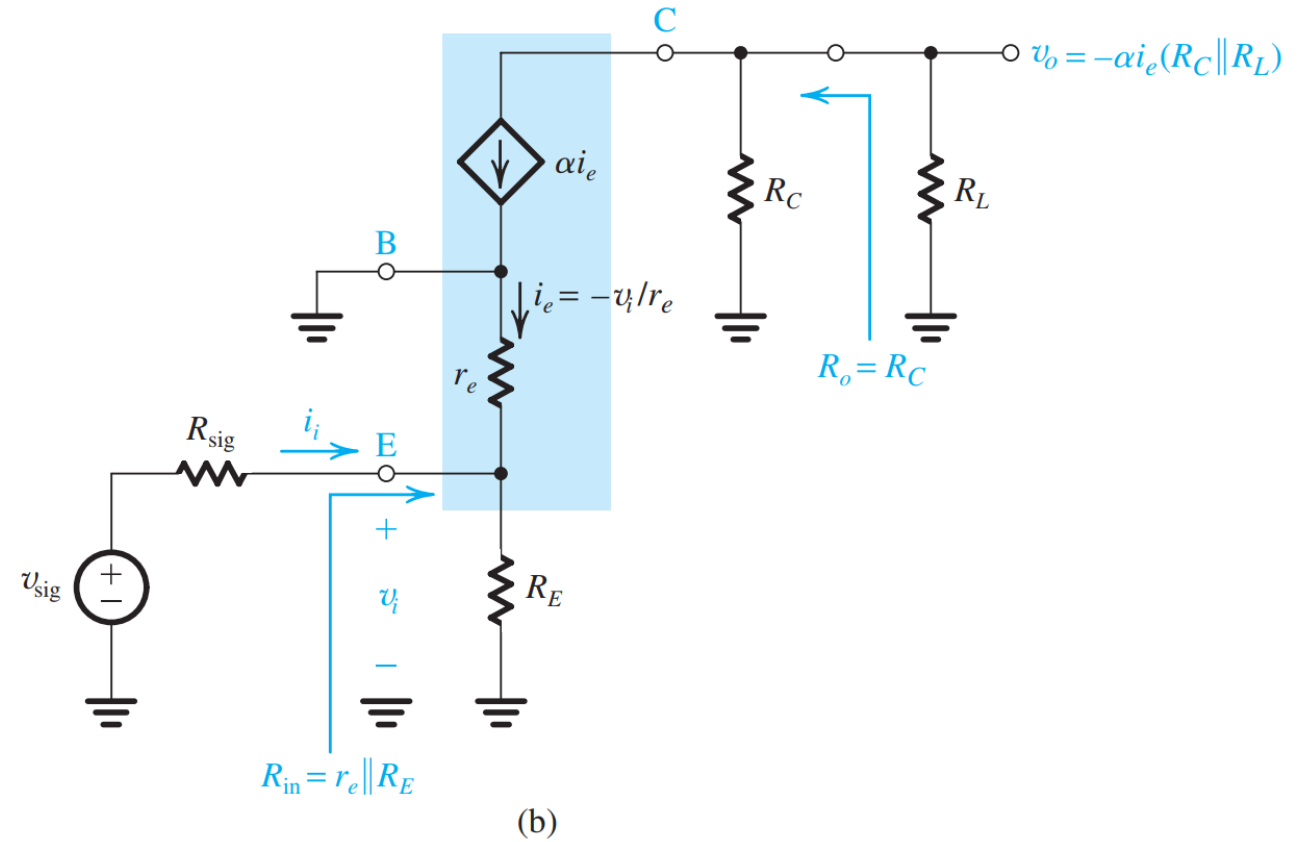
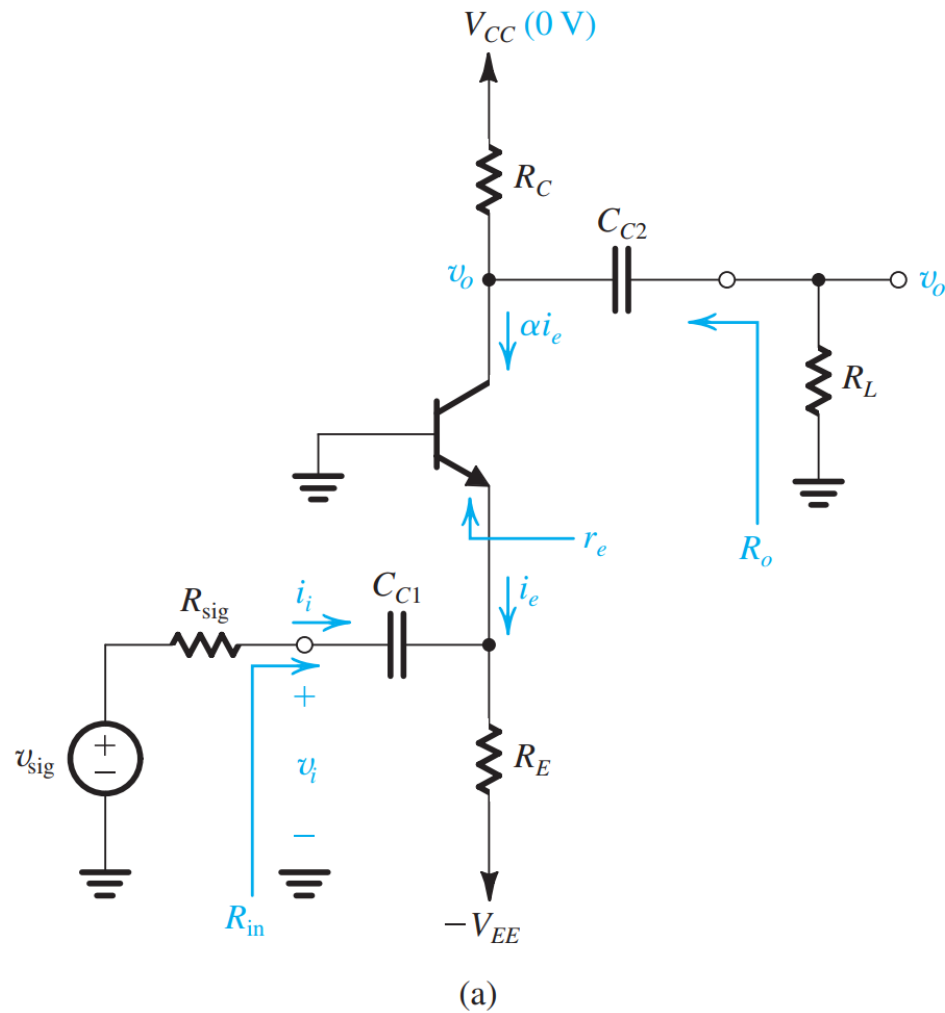
**Output Coupling Capacitor / DC Blocking Capacitor ( $C_{C2}$ ):** Placed between the transistor's collector (or emitter for some topologies) and the load.

- **Function:** Prevents the amplifier's DC collector voltage from damaging or altering the load, passing only the amplified AC signal.

**Emitter Bypass Capacitor ( $C_E$ ):** Connected in parallel with the emitter resistor ( $R_E$ ).

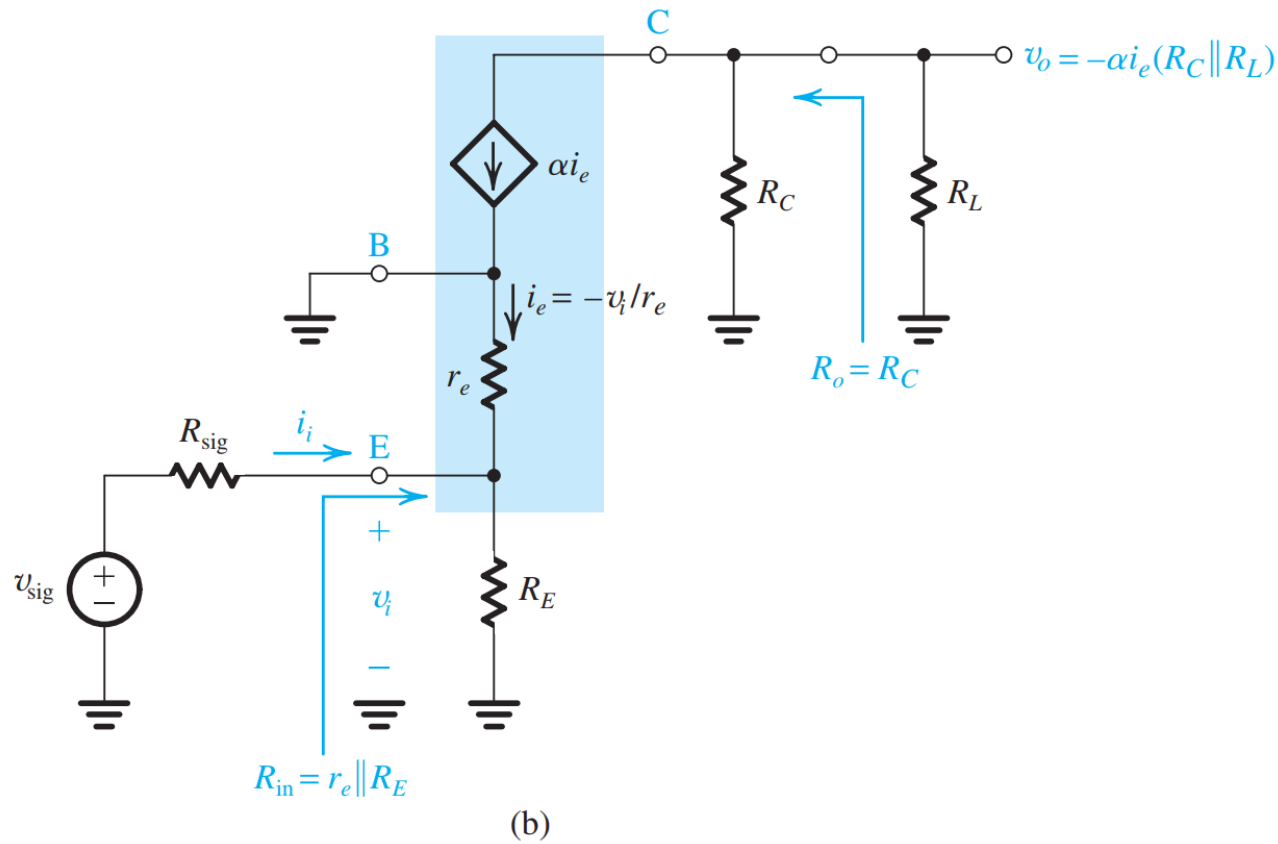
- **Function:** Provides a low-impedance path to ground for AC signals. By "bypassing" the emitter resistor at AC frequencies, it maximizes the AC voltage gain of the amplifier.

# The CB Amplifier (Bias Omitted)





# The CB Amplifier (Bias Omitted)



$$R_{in} = r_e \parallel R_E \simeq r_e \simeq 1/g_m$$

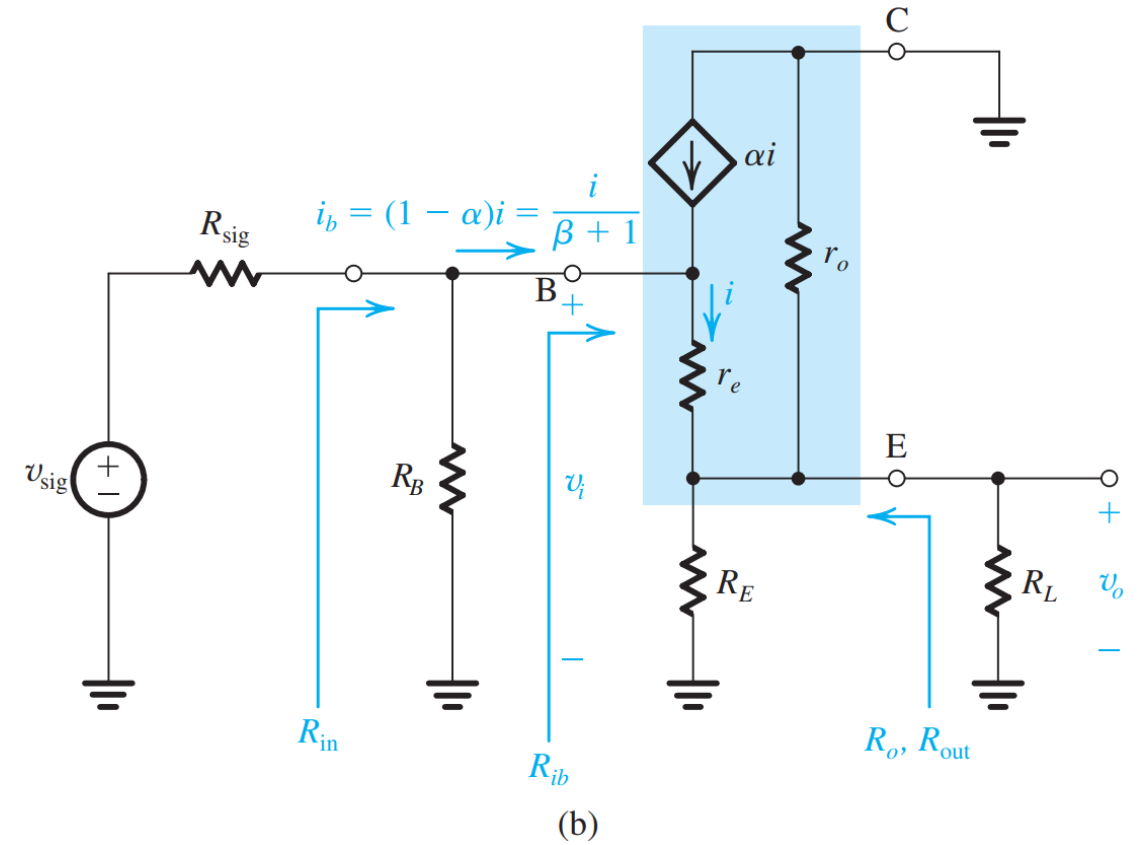
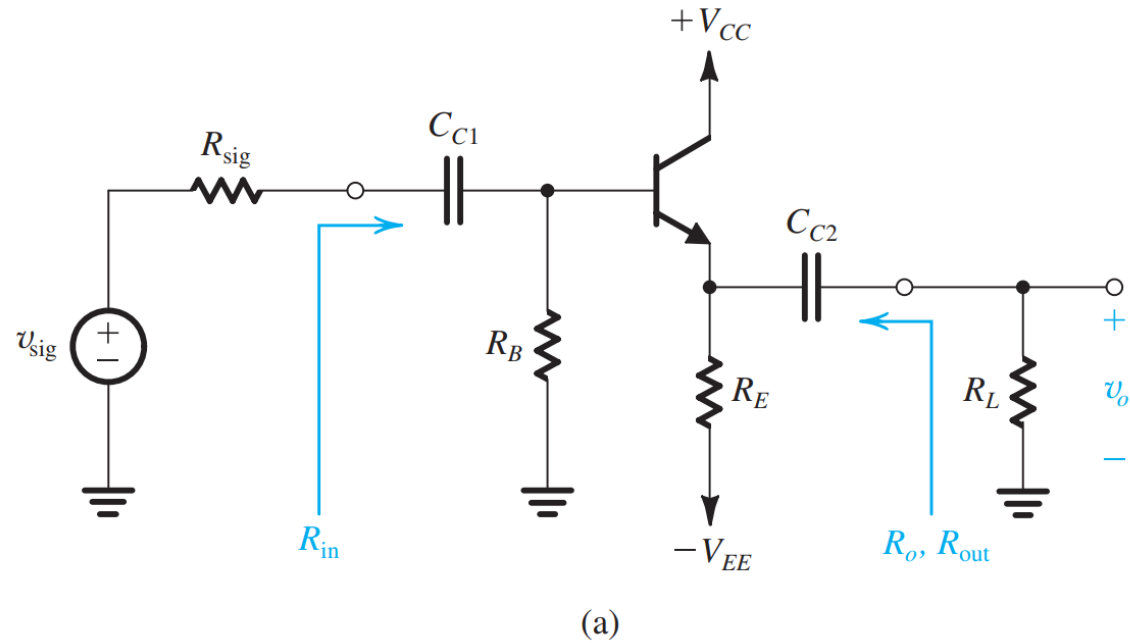
$$v_i = v_{sig} \frac{R_{in}}{R_{in} + R_{sig}}$$

$$i_e = -\frac{v_i}{r_e}$$

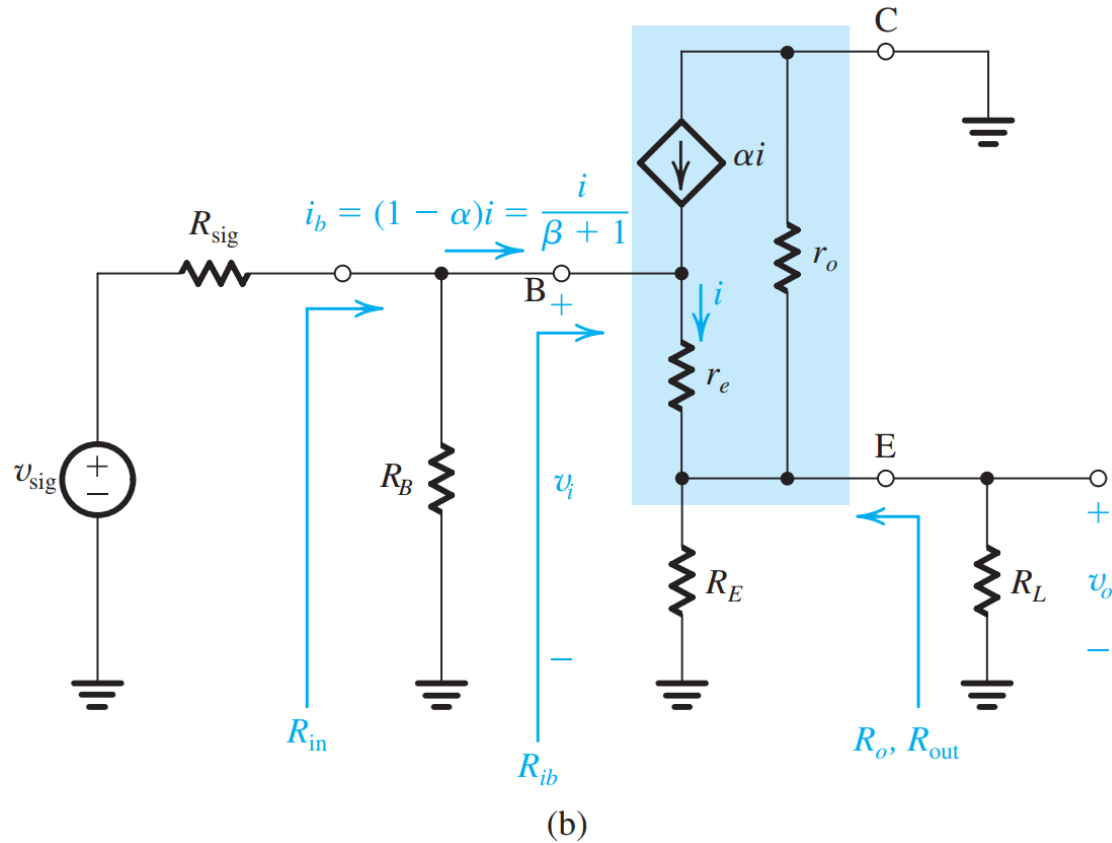
$$v_o = -\alpha i_e (R_C \parallel R_L)$$

$$G_v = \alpha \frac{R_{in}}{R_{in} + R_{sig}} \frac{R_C \parallel R_L}{r_e} = \frac{R_{in}}{R_{in} + R_{sig}} g_m (R_C \parallel R_L)$$

# The Emitter Follower (CC) Amplifier (Bias Omitted)



# The Emitter Follower (CC) Amplifier (Bias Omitted)



$$R_{in} = R_B \parallel R_{ib}$$

$$R_{ib} = (\beta + 1)[r_e + (R_E \parallel r_o \parallel R_L)]$$

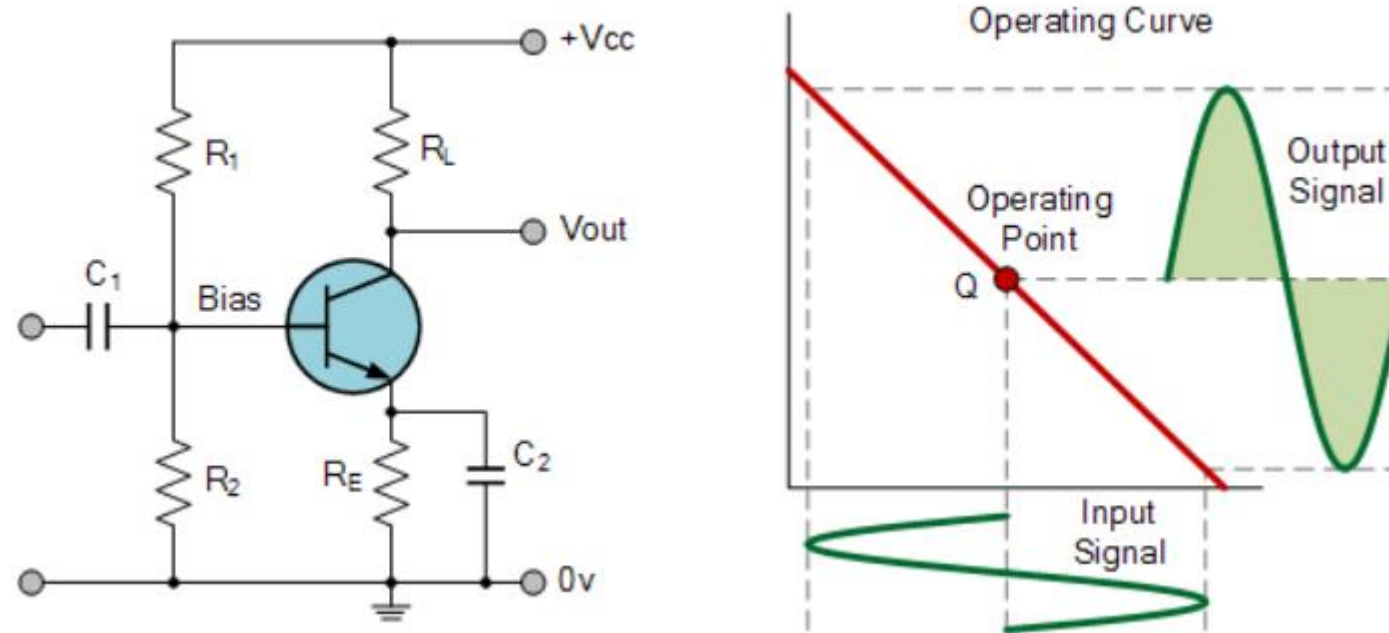
$$v_i = v_{sig} \frac{R_{in}}{R_{in} + R_{sig}}$$

$$v_o = v_i \frac{R_E \parallel r_o \parallel R_L}{r_e + (R_E \parallel r_o \parallel R_L)}$$

$$G_v \equiv \frac{v_o}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} \frac{(R_E \parallel r_o \parallel R_L)}{r_e + (R_E \parallel r_o \parallel R_L)}$$

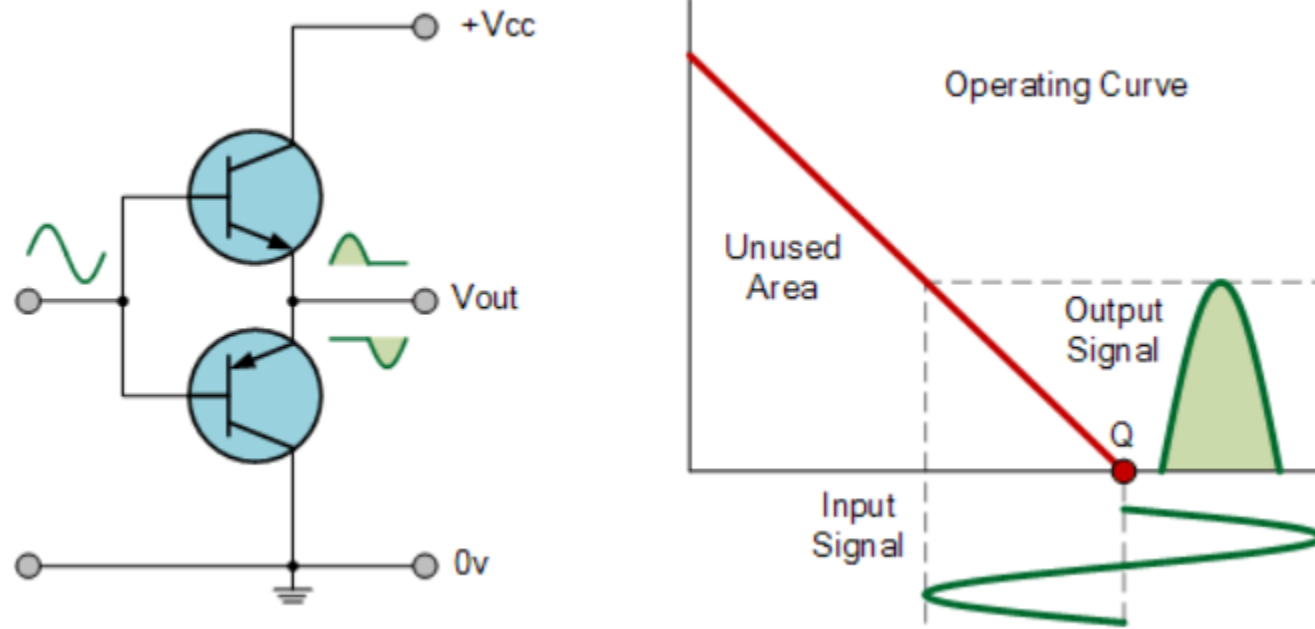
$$R_{out} = r_o \parallel R_E \parallel \left[ r_e + \frac{R_B \parallel R_{sig}}{\beta + 1} \right]$$

# The Push-Pull Amplifier (Class A)



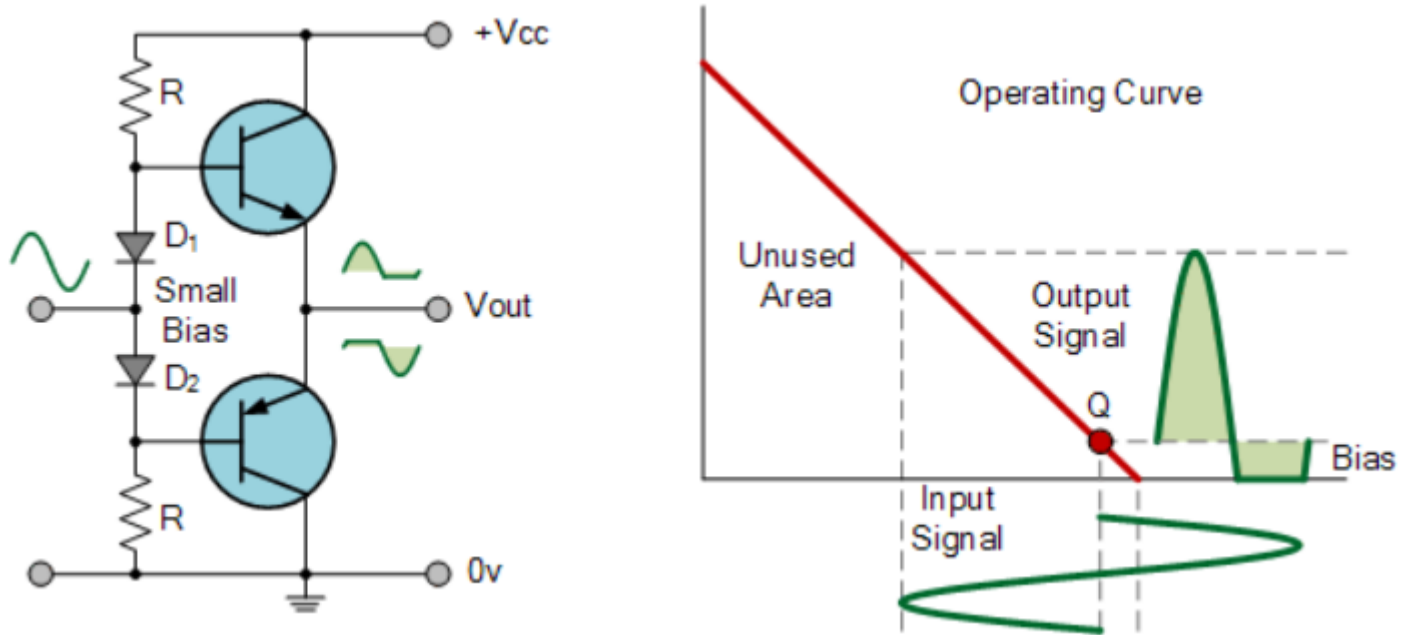
- ❑ High idling current represents a continuous loss of power.
- ❑ Very poor efficiency.

# The Push-Pull Amplifier (Class B)



- ❑ No idling current, better efficiency
- ❑ Crossover distortion, reduced linearity

# The Push-Pull Amplifier (Class AB)



□ A good compromise between efficiency and linearity